Chapter 11

Three-Dimensional Geometric and Modeling Transformations

Translation

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$P' = T \cdot P$$

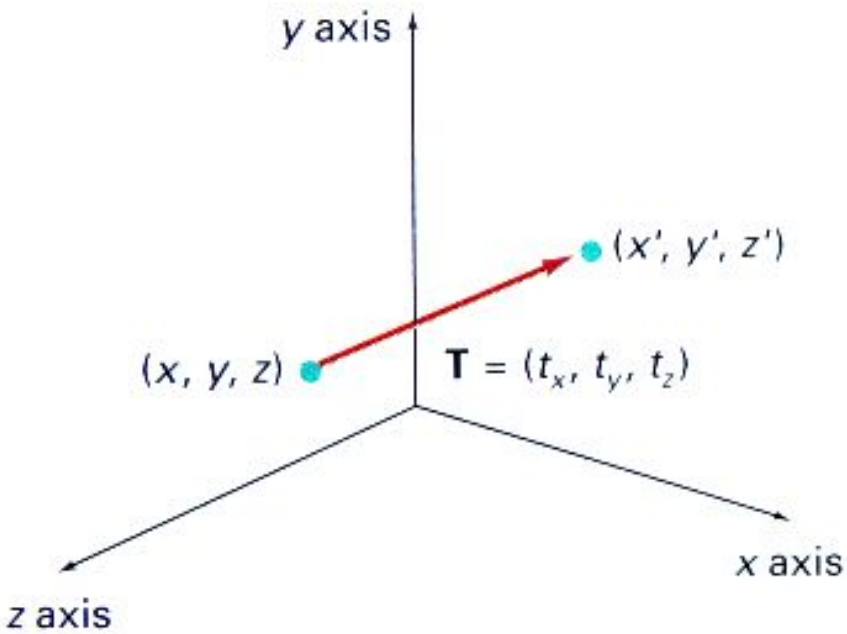
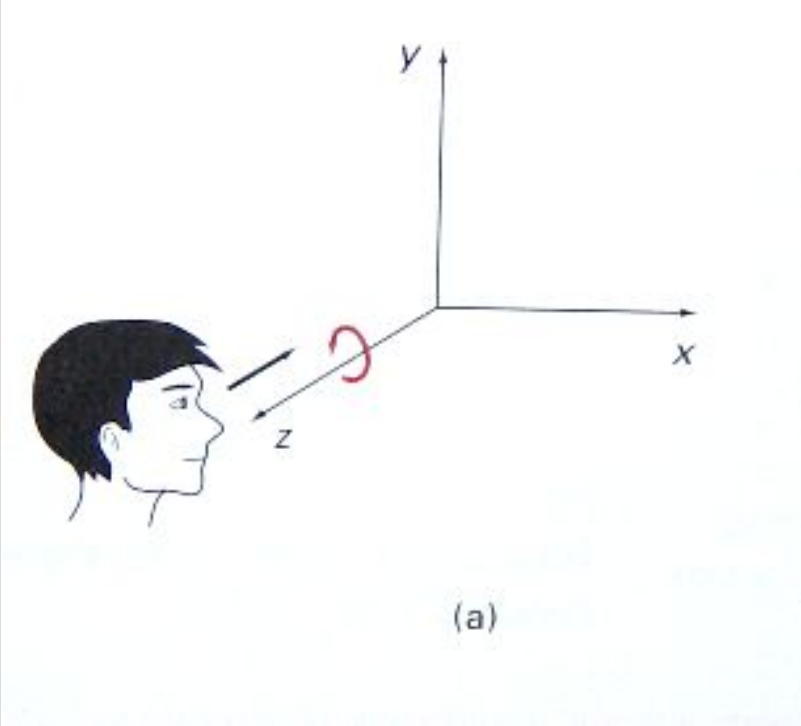
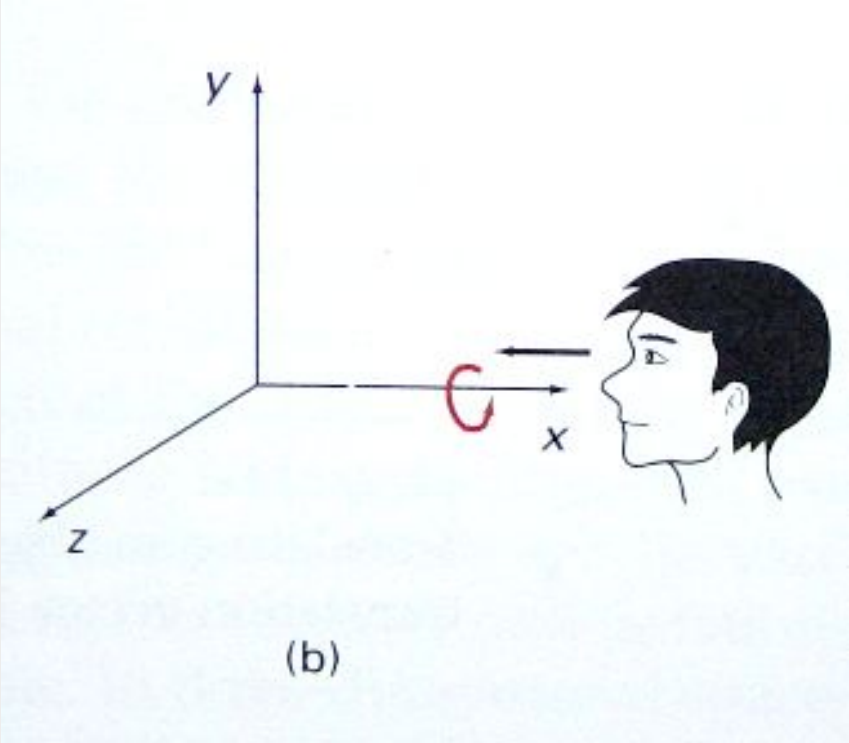


Figure 11-1
Translating a point with translation vector $\mathbf{T} = (t_x, t_y, t_z)$.

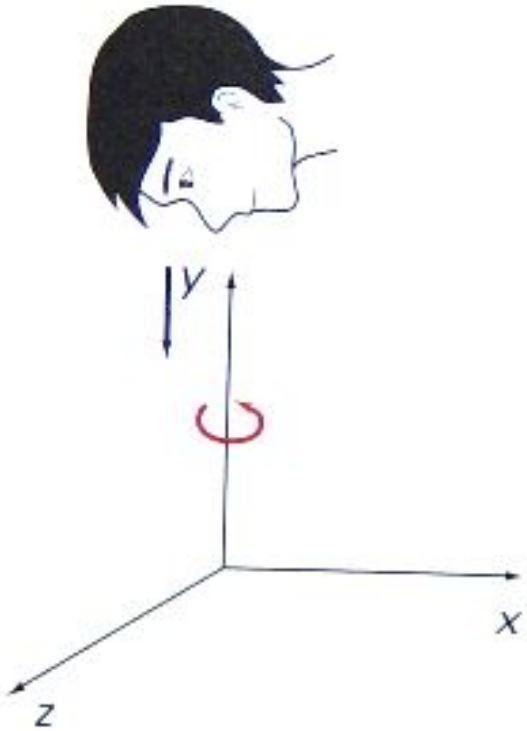
Rotation

Positive rotation angles produce counterclockwise rotations about a coordinate axis





Rotation



(c)

Figure 11-3
Positive rotation directions about the coordinate axes are counterclockwise, when looking toward the origin from a positive coordinate position on each axis.

Coordinate-Axes Rotations

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{cases} x' = x\cos\theta - y\sin\theta \\ y' = x\sin\theta + y\cos\theta \\ z' = z \end{cases}$$

$$P' = R_z(\theta)P$$

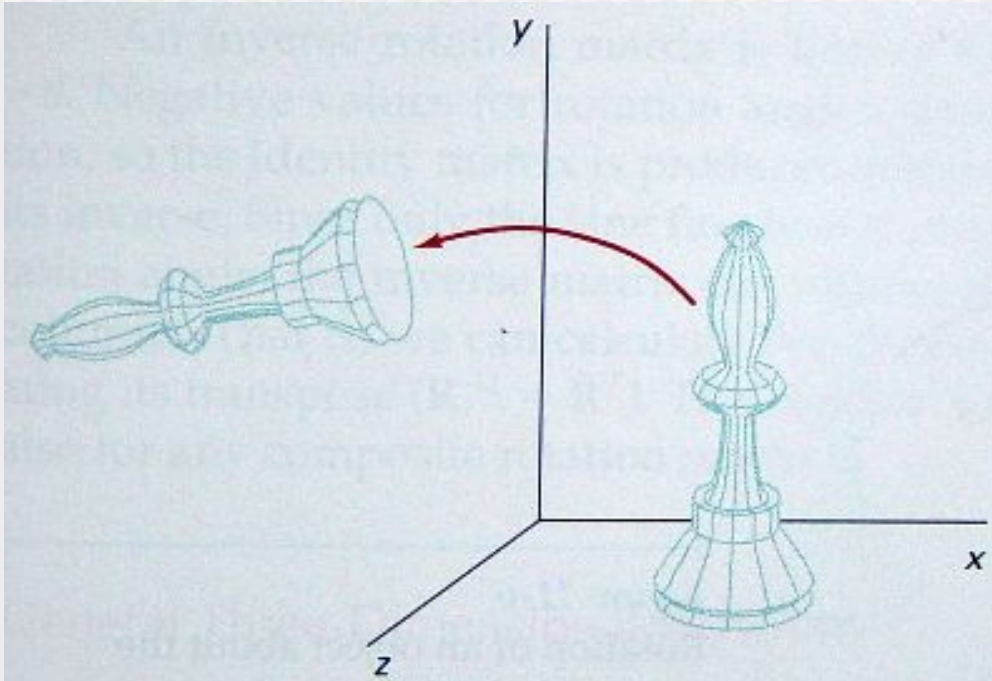


Figure 11-4
Rotation of an object about the z axis.

Coordinate-Axes Rotations

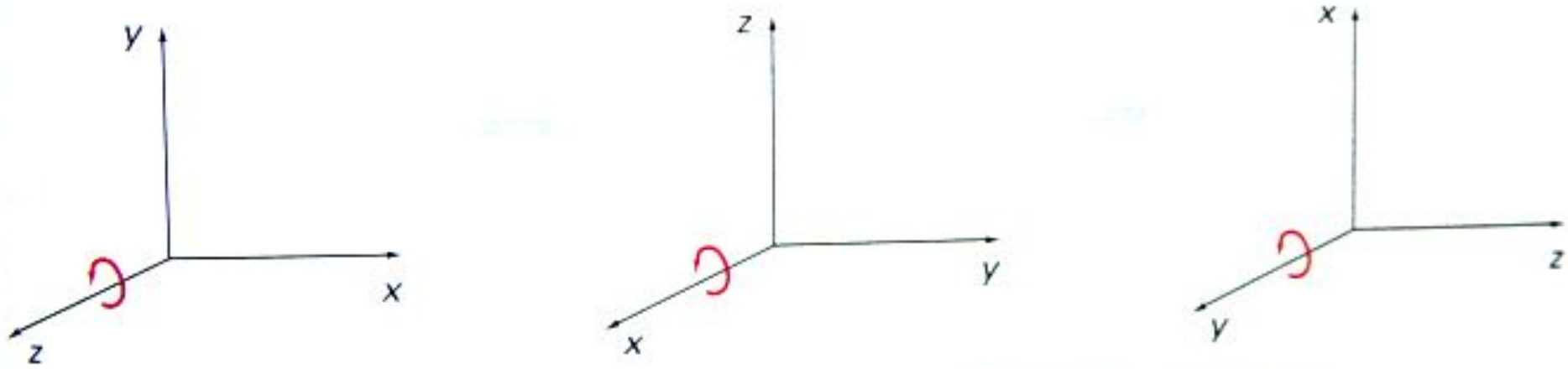


Figure 11-5
Cyclic permutation of the Cartesian-coordinate axes to produce the three sets of coordinate-axis rotation equations.

$$x \rightarrow y \rightarrow z \rightarrow x$$

Coordinate-Axes Rotations

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & -in\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \begin{cases} x' = x \\ y' = y\sin\theta + z\cos\theta \\ z' = y\cos\theta - z\sin\theta \end{cases}$$

$$P' = R_x(\theta)P$$

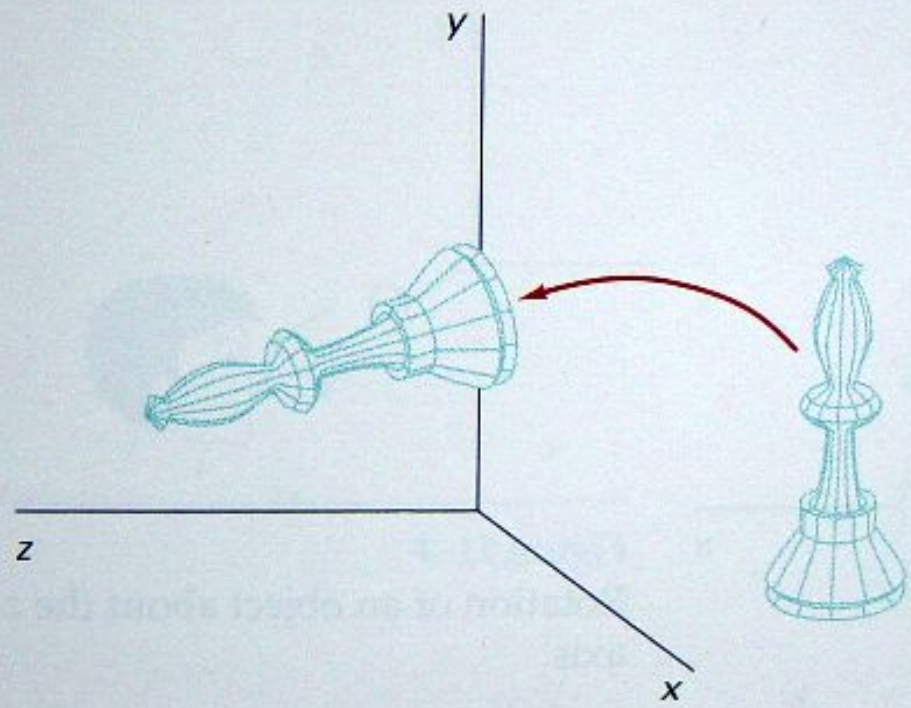


Figure 11-6
Rotation of an object about the x axis.

Coordinate-Axes Rotations

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{cases} x' = z\sin\theta + x\cos\theta \\ y' = y \\ z' = z\cos\theta - x\sin\theta \end{cases}$$

$$P' = R_y(\theta)P$$

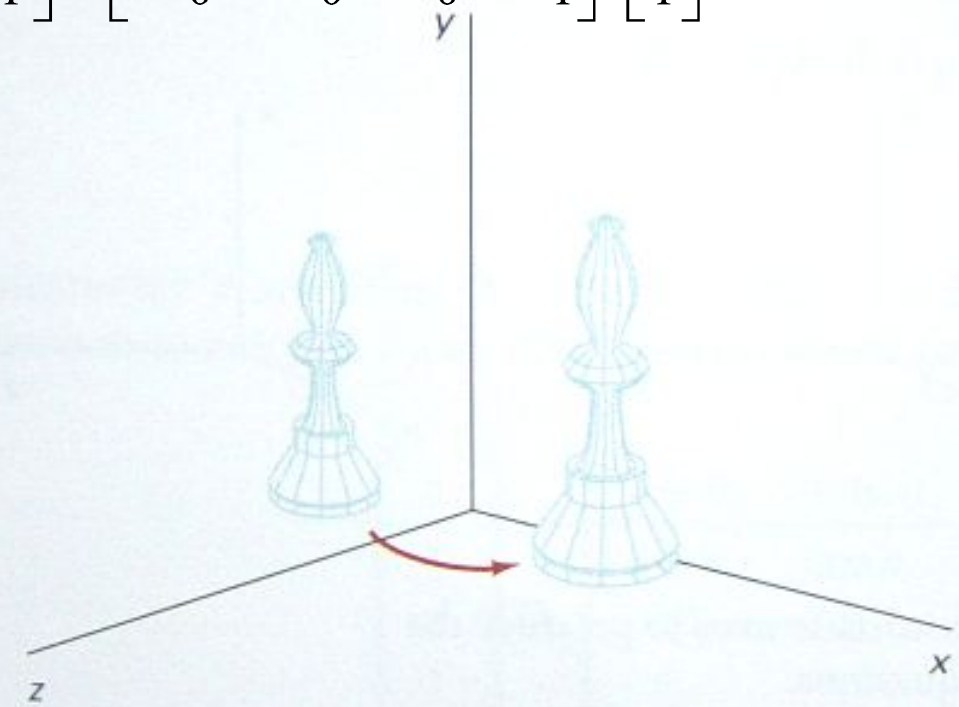


Figure 11-7
Rotation of an object about the *y* axis.

G
R

- s

axis
s
axis
axis

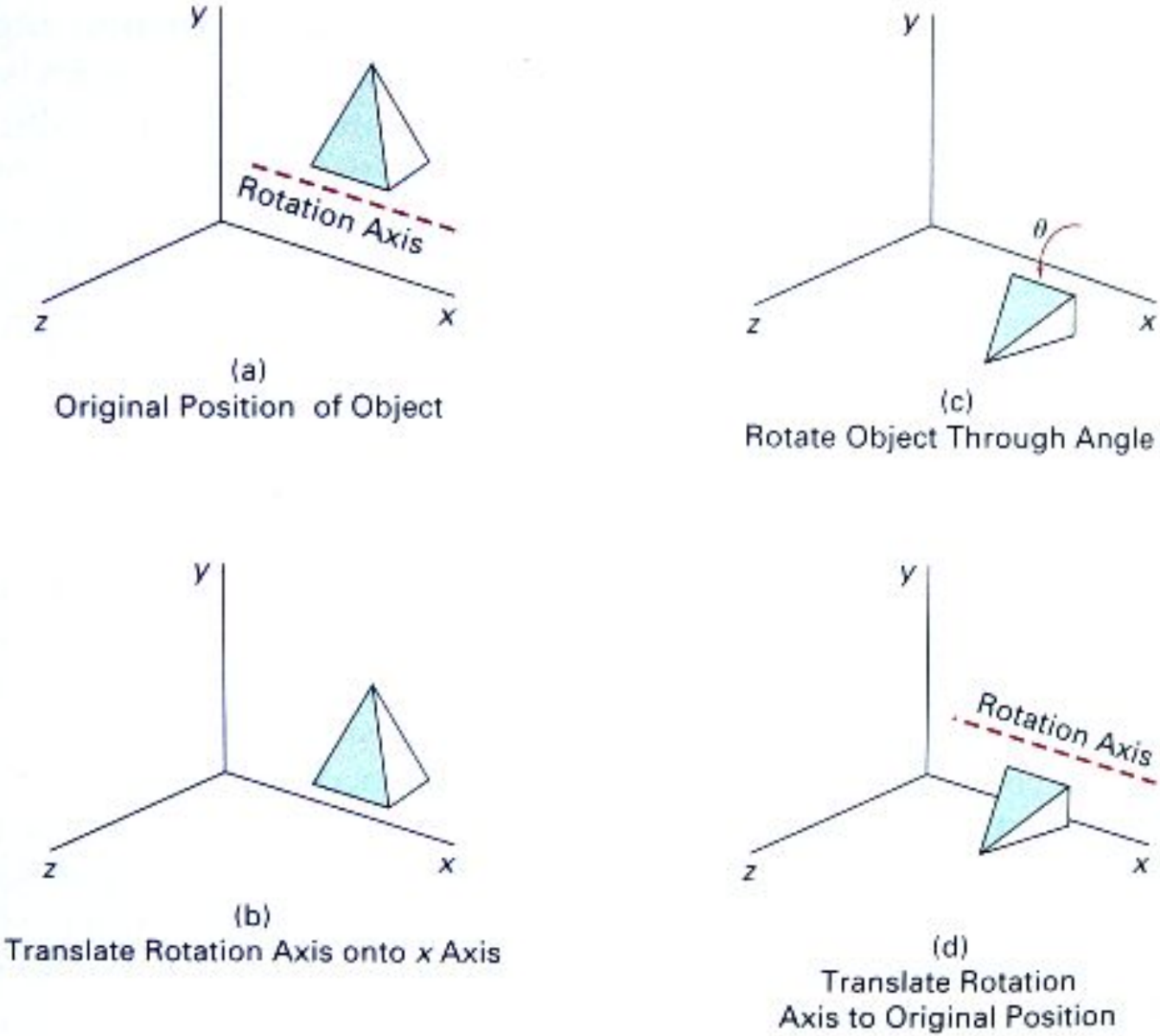


Figure 11-8
Sequence of transformations for rotating an object about an axis that is parallel to the x axis.

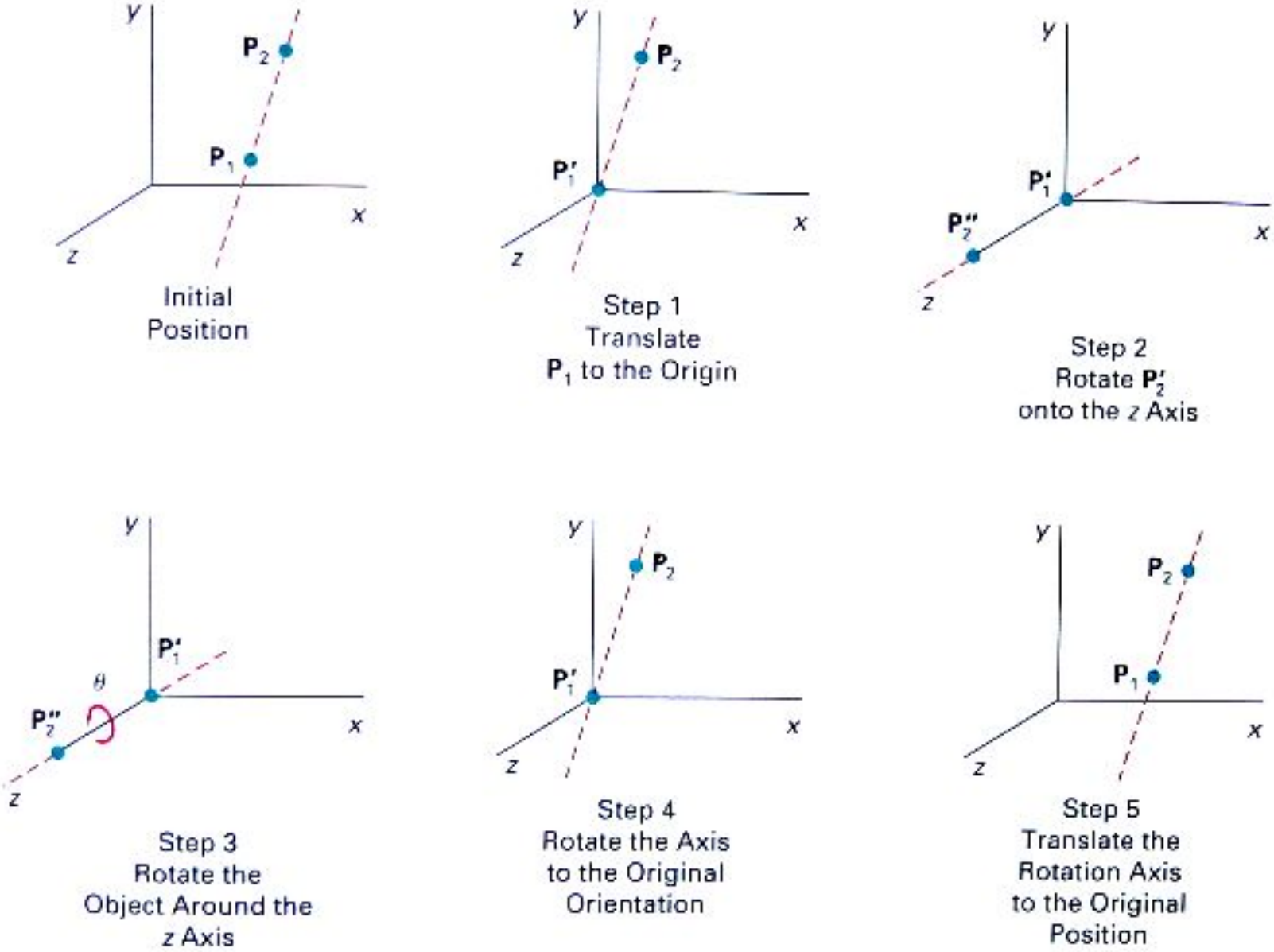


Figure 11-9
Five transformation steps for obtaining a composite matrix for rotation about an arbitrary axis, with the rotation axis projected onto the z axis.

General Three-Dimensional Rotations

- **Transforming the rotation axis onto the z-axis.**
- Rotation axis vector V=P2-P1 =(x2-x1,y2-y1,z2-z1)

Unit vector *u* is defined along the rotation axis as:

$$u = \frac{V}{|V|} = (a,b,c)$$

where *a,b,c* are the direction cosines for the rotation axis:

$$a = \frac{x_2 - x_1}{|V|} \quad b = \frac{y_2 - y_1}{|V|} \quad c = \frac{z_2 - z_1}{|V|}$$

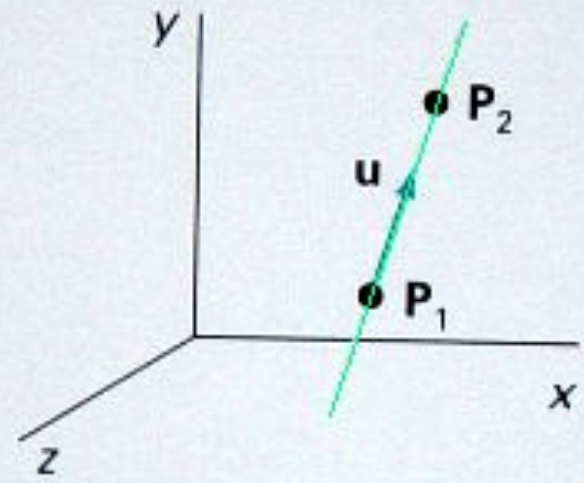


Figure 11-10
An axis of rotation (dashed line) defined with points $\mathbf{P}_1$ and $\mathbf{P}_2$. The direction for the unit axis vector **u** is determined by the specified rotation direction.

Step1:set up the translation matrix

$$T = \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

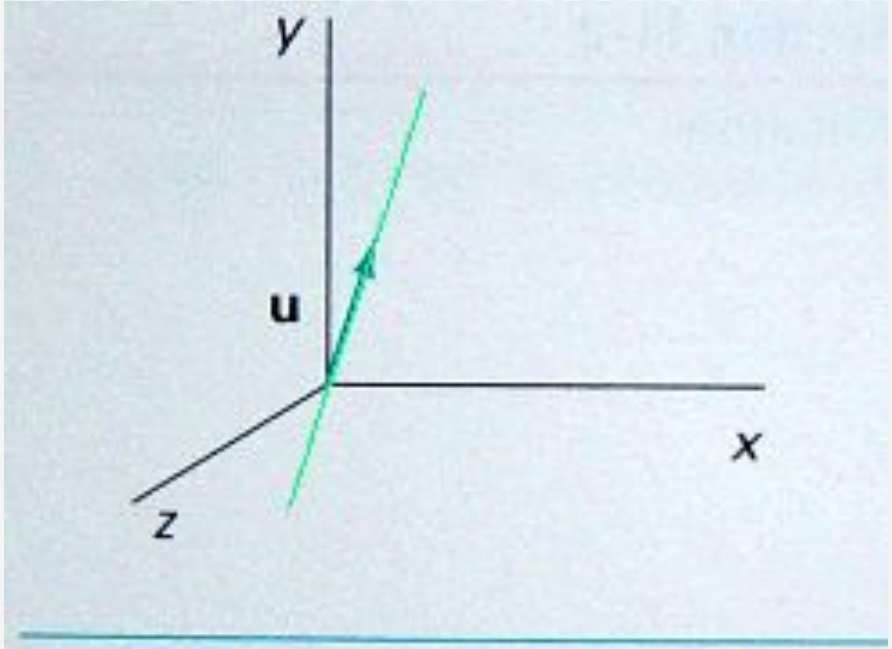


Figure 11-11
Translation of the rotation axis to the coordinate origin.

Step2:put the rotation axis on the z axis

1. Rotate about the x axis to transform vector u into xz plane
2. Swing u around to the z axis using a y-axis rotation

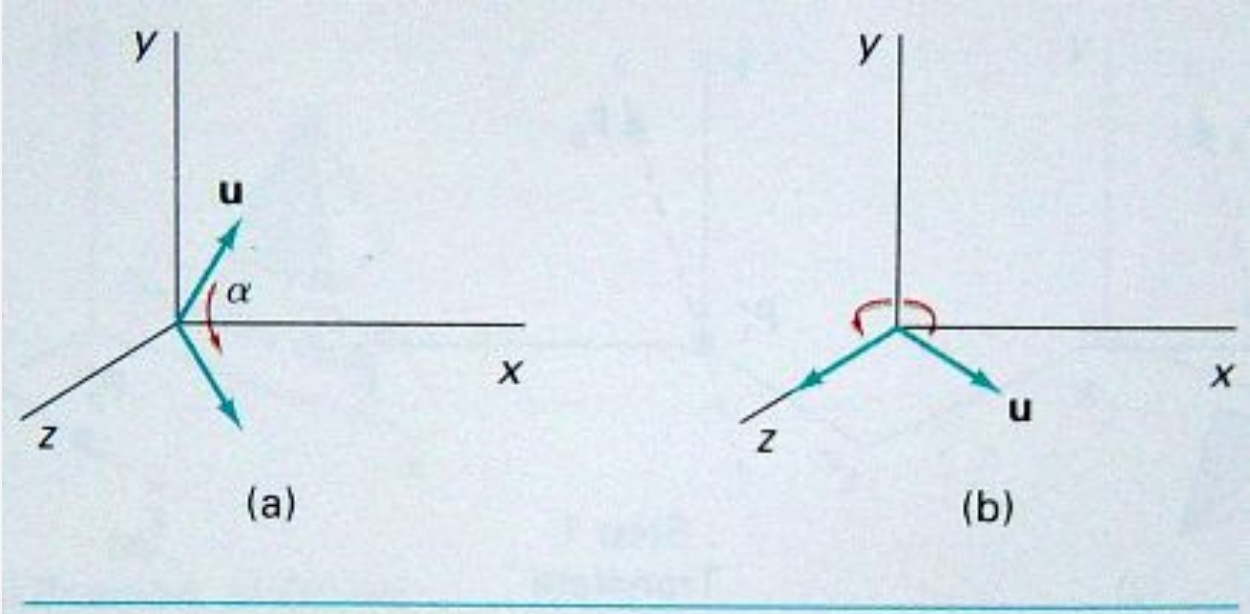


Figure 11-12
Unit vector **u** is rotated about the x axis to bring it into the xz plane (a), then it is rotated around the y axis to align it with the z axis (b).

transform vector u into xz plane

Projection of *u* in the *yz* plane be *u'* & *d* be the magnitude of *u'*

$$u' = (0, b, c) \qquad u_z = (0,0,1)$$

$$\cos\alpha = \frac{u' \cdot u_z}{|u'||u_z|} = \frac{c}{d} \qquad d = \sqrt{b^2 + c^2}$$

Coordinate independent form:

$$u' \times u_z = u_x |u'||u_z| \sin\alpha \qquad \text{(i)}$$

Cartesian form:

$$u' \times u_z = \begin{vmatrix} u_x & u_y & u_z \\ 0 & b & c \\ 0 & 0 & 1 \end{vmatrix} = u_x \cdot b \qquad \text{(ii)}$$

(i) & (ii) =>

$$d \sin\alpha = b \qquad \sin\alpha = \frac{b}{d}$$

where $|u_z|$=1 & |u'|=d

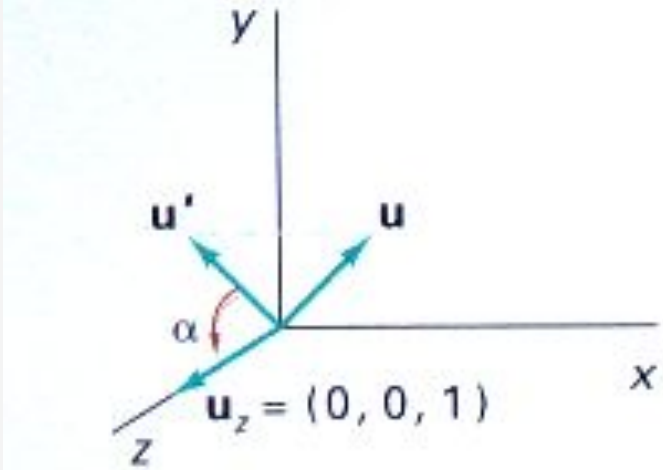


Figure 11-13
Rotation of **u** around the *x* axis into the *xz* plane is accomplished by rotating **u'** (the projection of **u** in the *yz* plane) through angle *α* onto the z axis.

$$R_z(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c/d & -b/d & 0 \\ 0 & b/d & c/d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This matrix rotates unit vector *u* about the *x* –axis into the *xz* plane

Swing u around to the z axis using a y-axis rotation

$$u'' = (a, 0, d)$$

It's z-component is d i.e., the magnitude of u' because vector u' has been rotated onto the z-axis & y-component =0 since it is in xz plane

$$\cos\beta = \frac{u''\cdot u_z}{|u''||u_z|} = d$$

$|u''| = |u_z| = 1$

$$u''\times u_z = u_y |u''||u_z| \sin\beta$$

$$u''\times u_z = u_y \cdot (-a)$$

$$\sin\beta = -a$$

$$R_y(\beta) = \begin{bmatrix} d & 0 & -a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

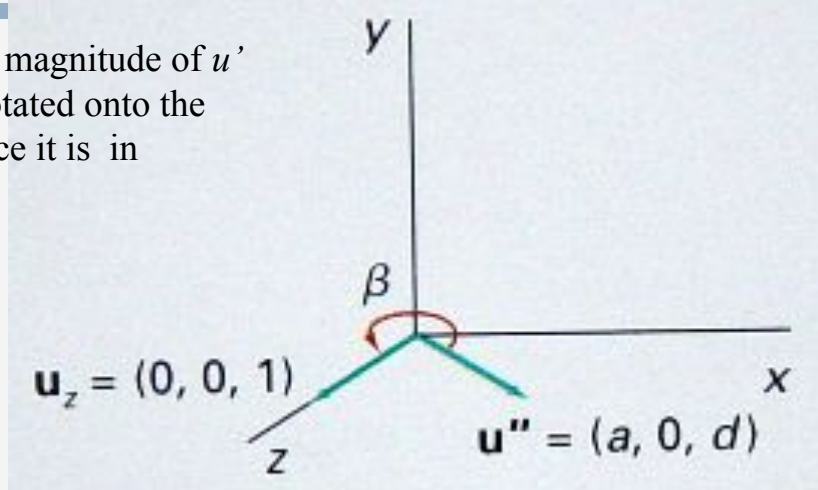


Figure 11-14
Rotation of unit vector $\mathbf{u}''$ (vector $\mathbf{u}$ after rotation into the xz plane) about the y axis. Positive rotation angle β aligns $\mathbf{u}''$ with vector $\mathbf{u}_z$.

Step3: rotation about the z axis θ

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Composition of transformations:

$$R(\theta) = T^{-1} \cdot R_x^{-1}(\alpha) \cdot R_y^{-1}(\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \cdot T$$

Scaling

- **Scaling with respect to the coordinate origin**

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$P' = S \cdot P$$

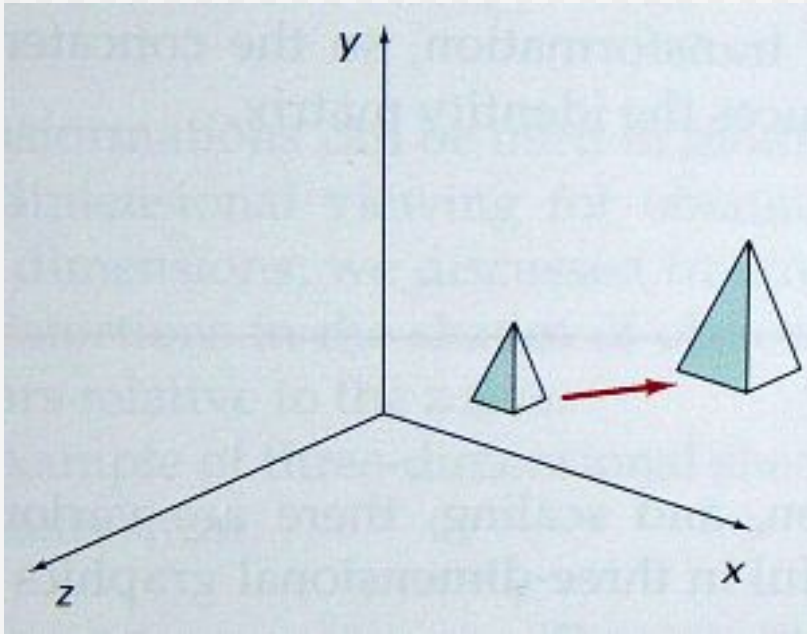


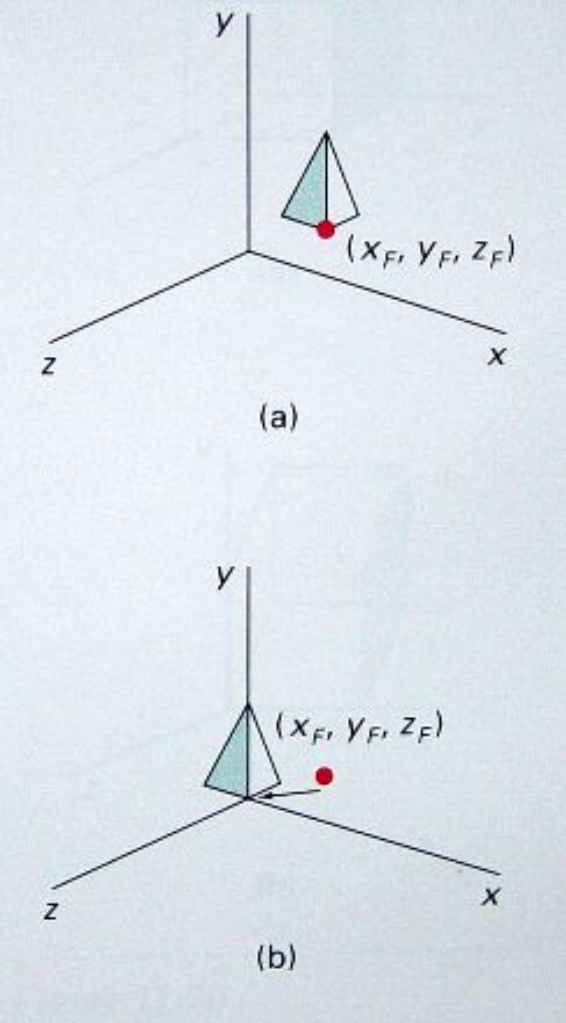
Figure 11-17
Doubling the size of an object with transformation 11-42 also moves the object farther from the origin.

Scaling

- **Scaling with respect to a selected fixed position (x_f, y_f, z_f)**
  1. Translate the fixed point to origin
  2. Scale the object relative to the coordinate origin
  3. Translate the fixed point back to its original position

$$T(x_f, y_f, z_f) \cdot S(s_x, s_y, s_z) \cdot T(-x_f, -y_f, -z_f) = \begin{bmatrix} s_x & 0 & 0 & (1-s_x)x_f \\ 0 & s_y & 0 & (1-s_y)y_f \\ 0 & 0 & s_z & (1-s_z)z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Scaling



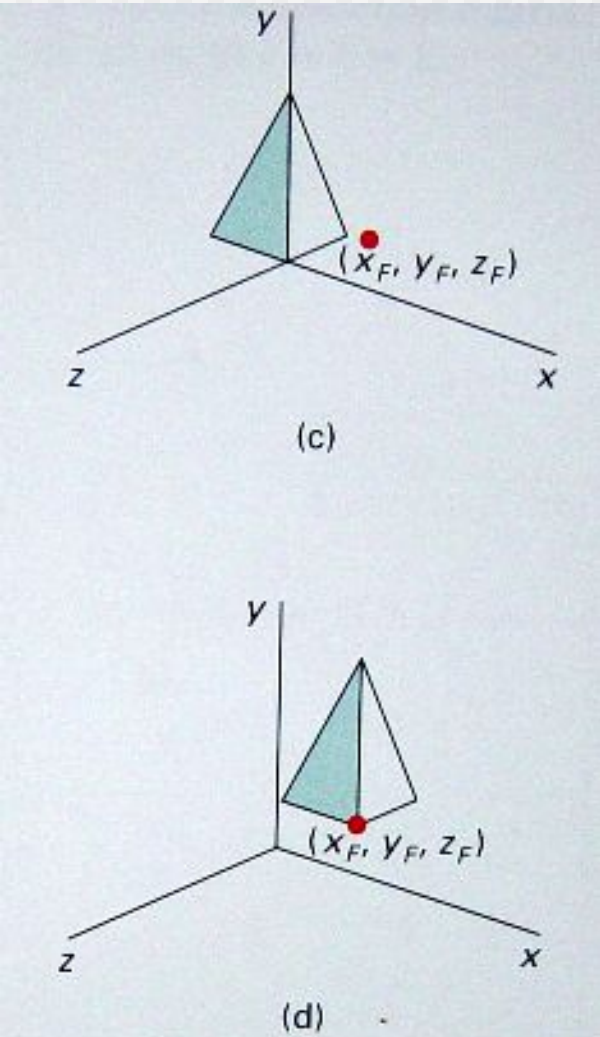


Figure 11-18
Scaling an object relative to a selected fixed point is equivalent to the sequence of transformations shown.

Reflections

$$RF_z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

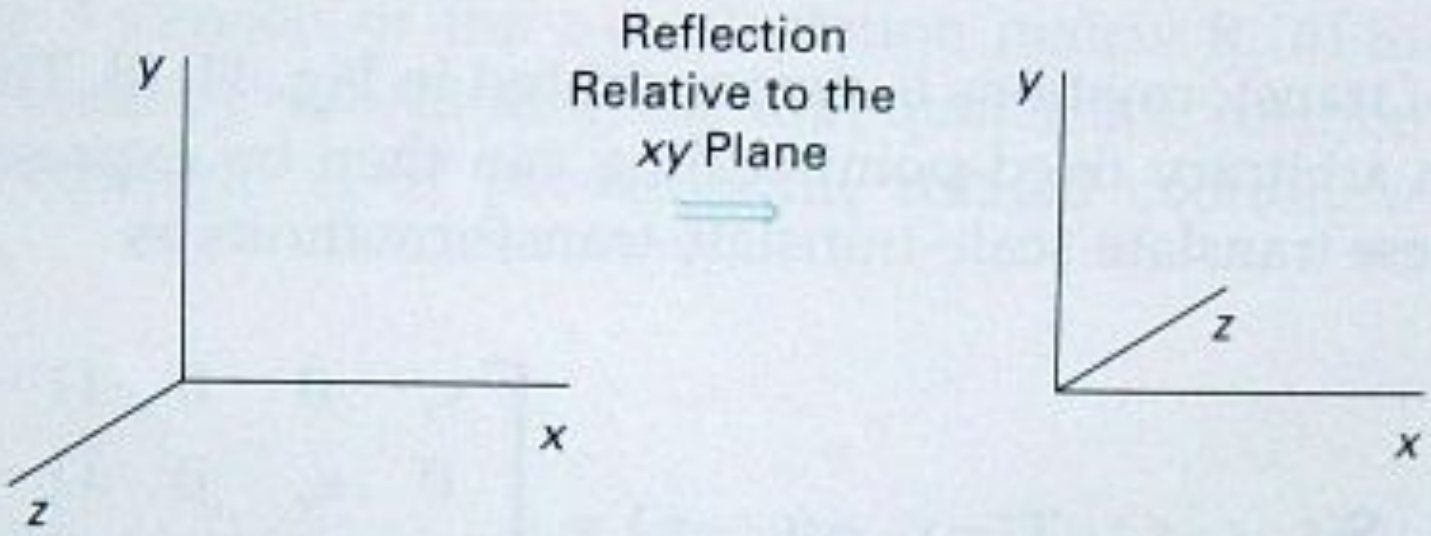


Figure 11-19
Conversion of coordinate specifications from a right-handed to a left-handed system can be carried out with the reflection transformation 11-46.

Shears

$$SH_z = \begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

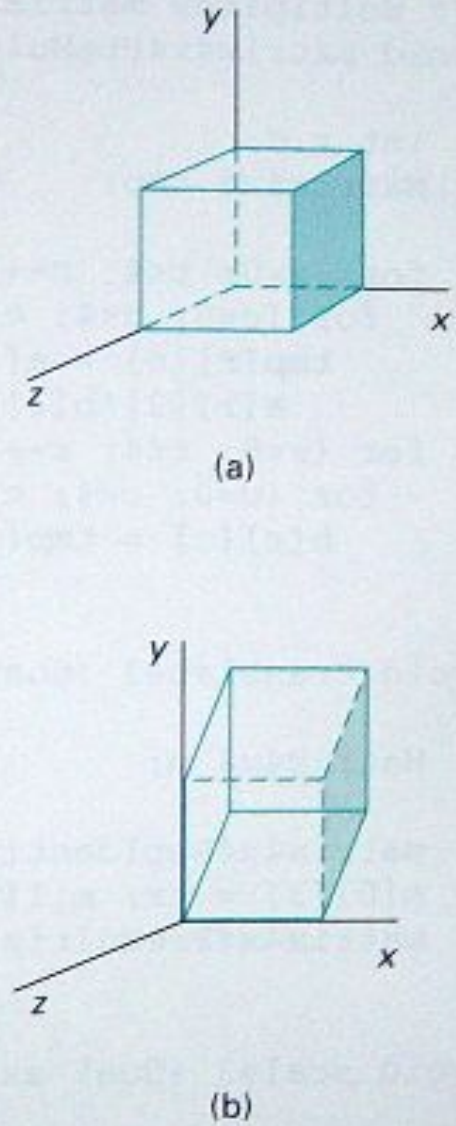


Figure 11-20
A unit cube (a) is sheared (b) by transformation matrix 11-47, with $a = b = 1$.

Composite transformations

Carried out from right to left, where the rightmost matrix is the first transformation to be applied to an object and the leftmost matrix is the last transformation.

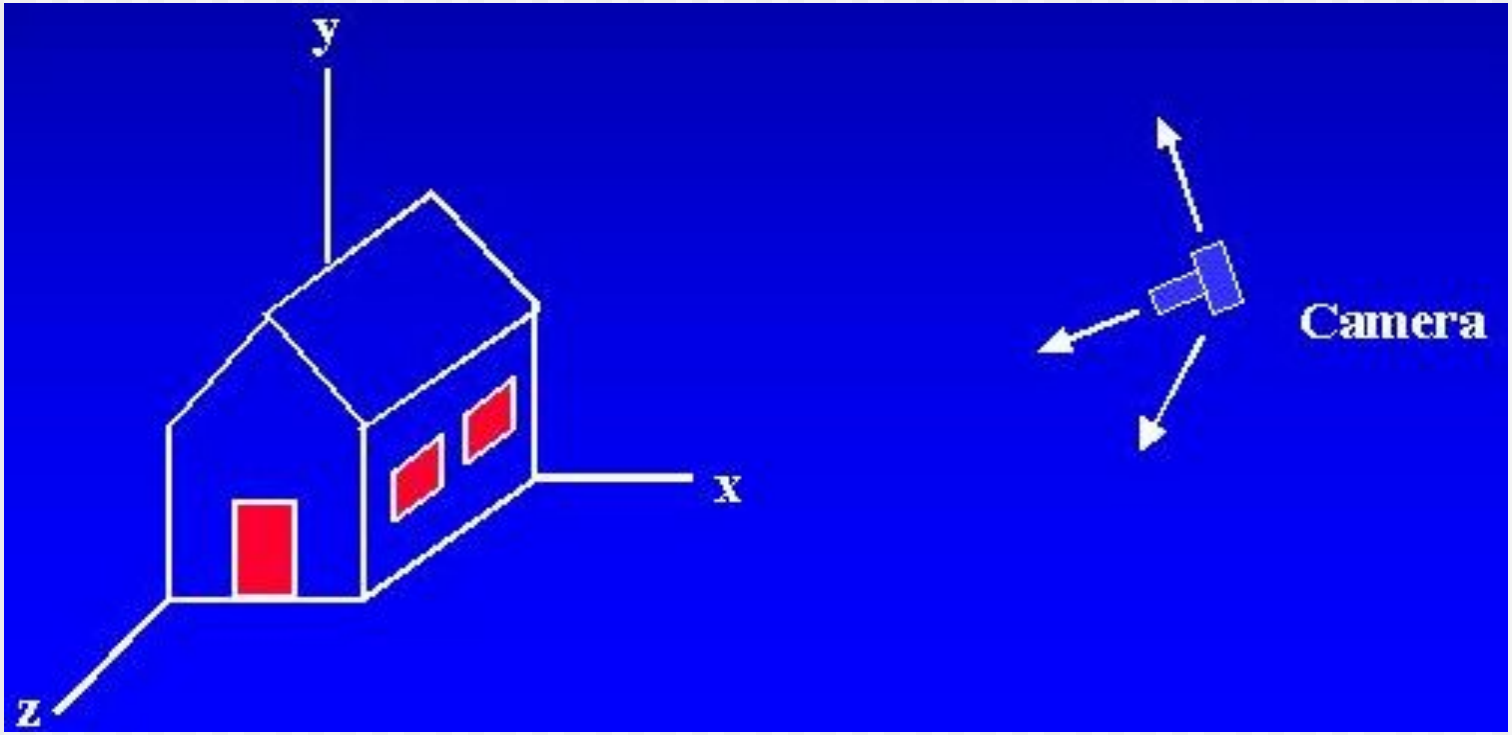
- Find the matrix for rotating any object by 30^0 about an axis passing through the origin and point (10,0,10).

Chapter 12

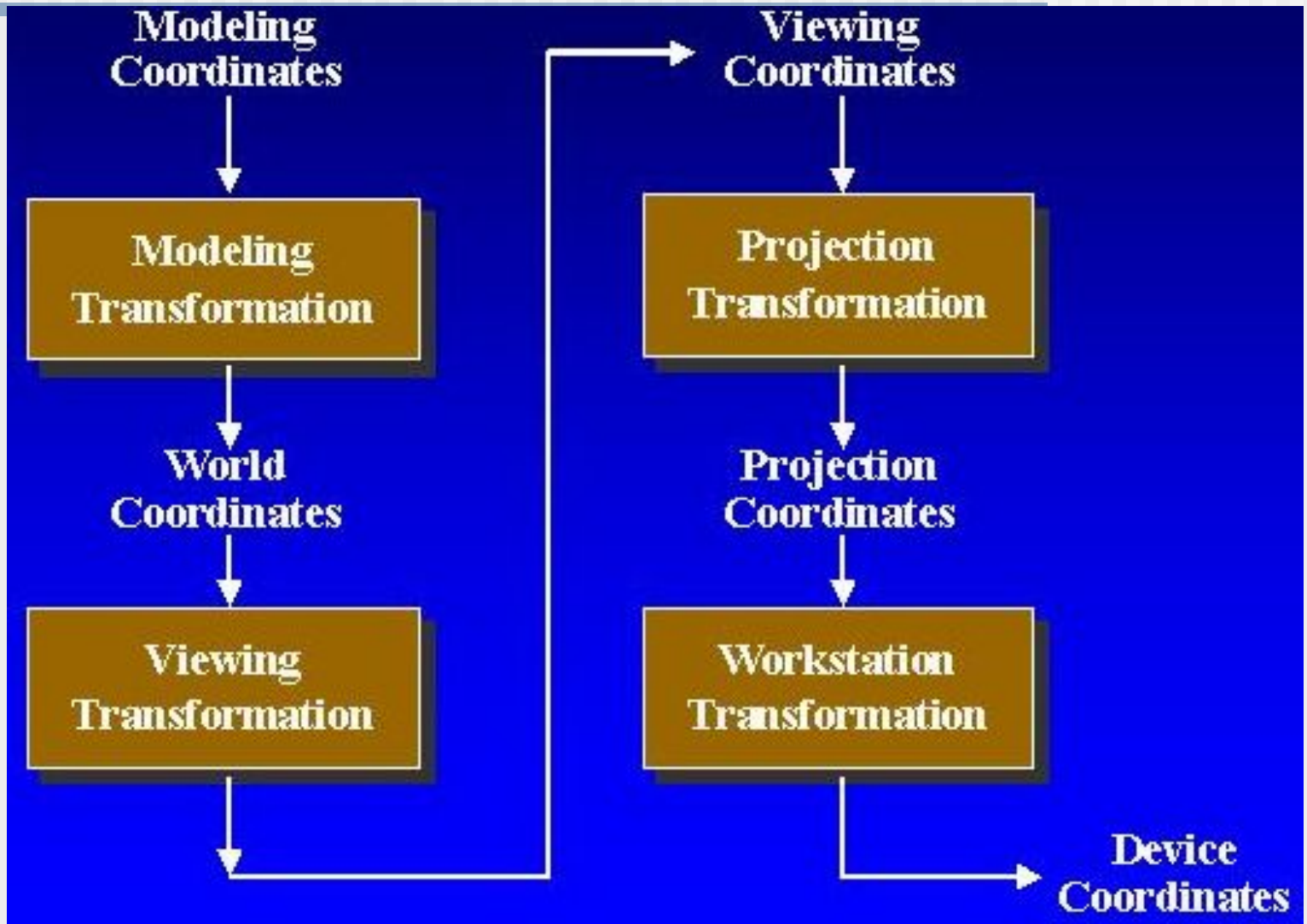
Three-Dimensional Viewing

Viewing

- Analogous to the photographing process
  - Camera position
  - Camera orientation



Viewing Pipeline



THE VIEWING SYSTEM

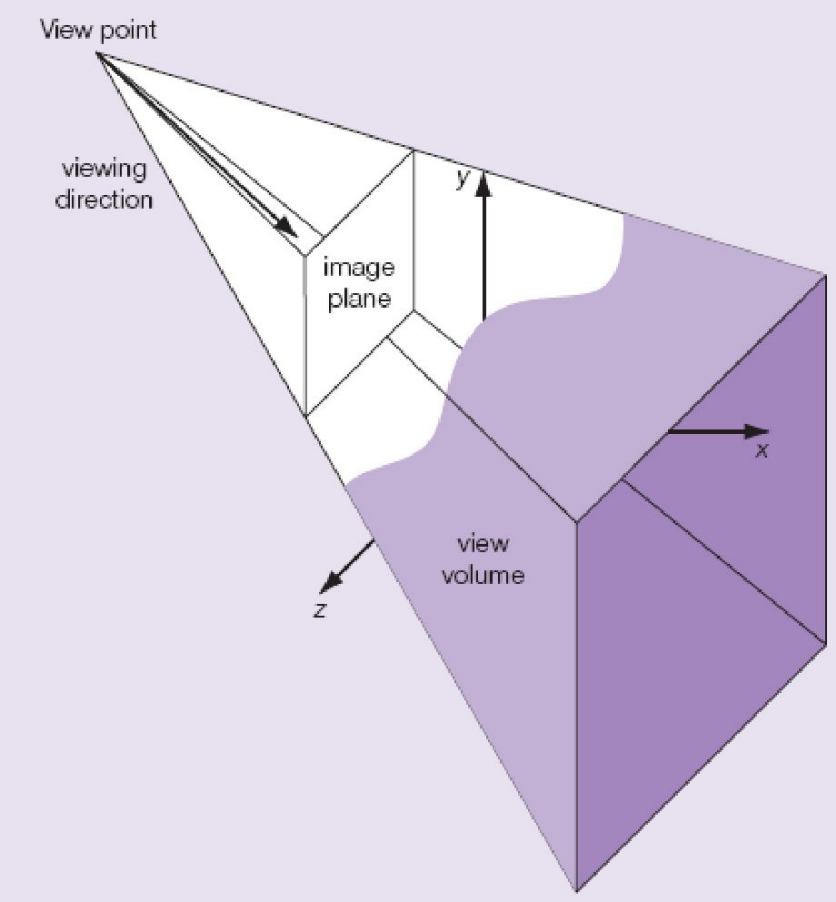
- The viewing system is a unified model for image visualization and consists of a view coordinate system and point of view.
- These two components establish the viewer's position in terms of world coordinates.
- The coordinate system is specified with respect to this point of view.

THE VIEWING SYSTEM

- The point of view can either be the origin of the coordinate system, or the center of projection.
- The viewing system must also contain an image plane for the projection of scenes and a view frustum/volume for the specification of the field of view.

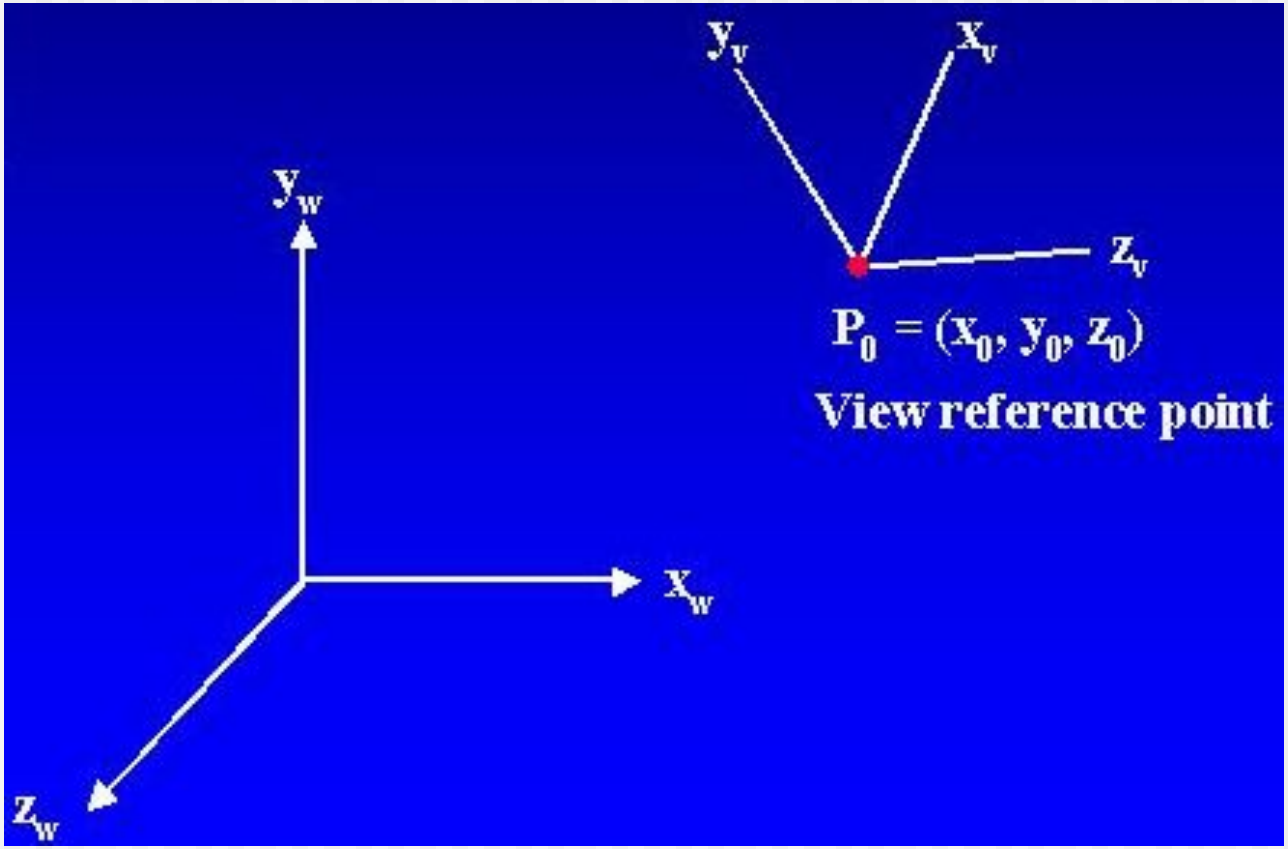
THE VIEWING SYSTEM

FIGURE 7-8 The viewing system



Viewing-Coordinate System

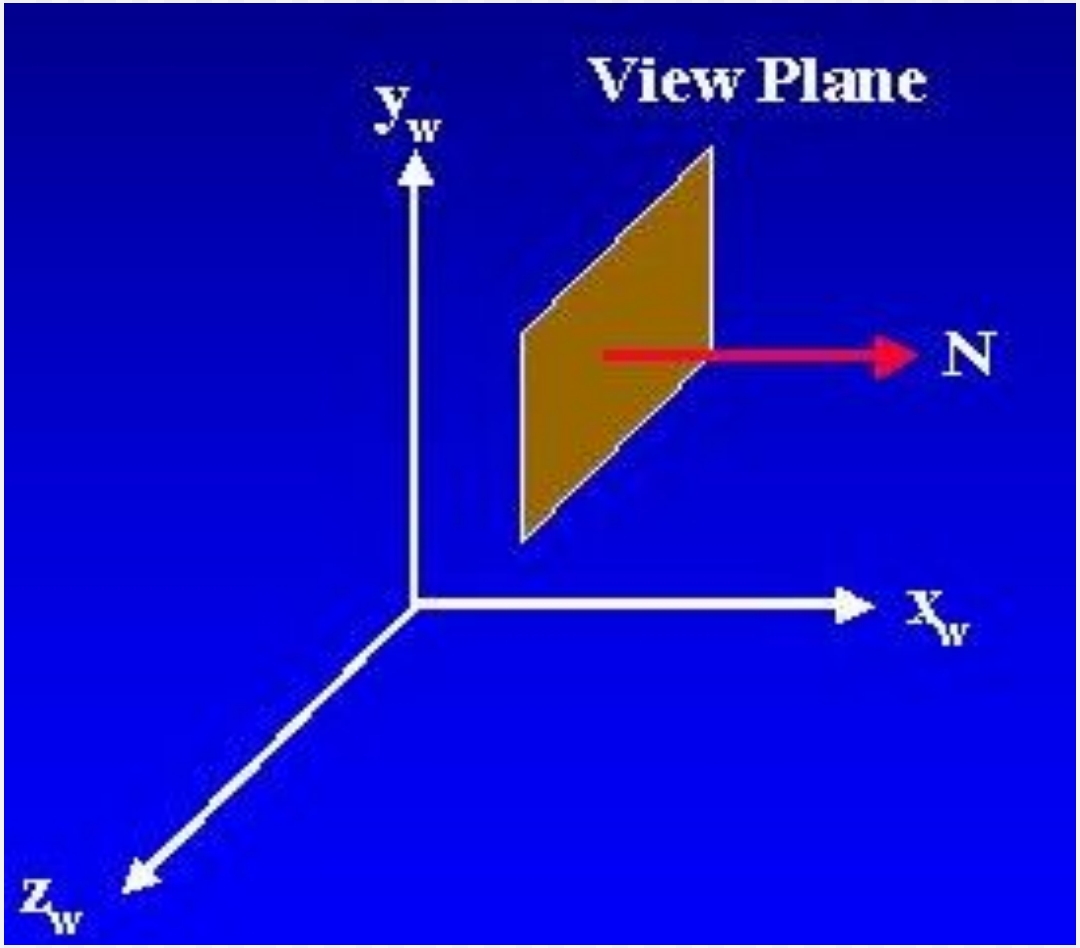
Viewing -Coordinate System or View Reference Coordinate System



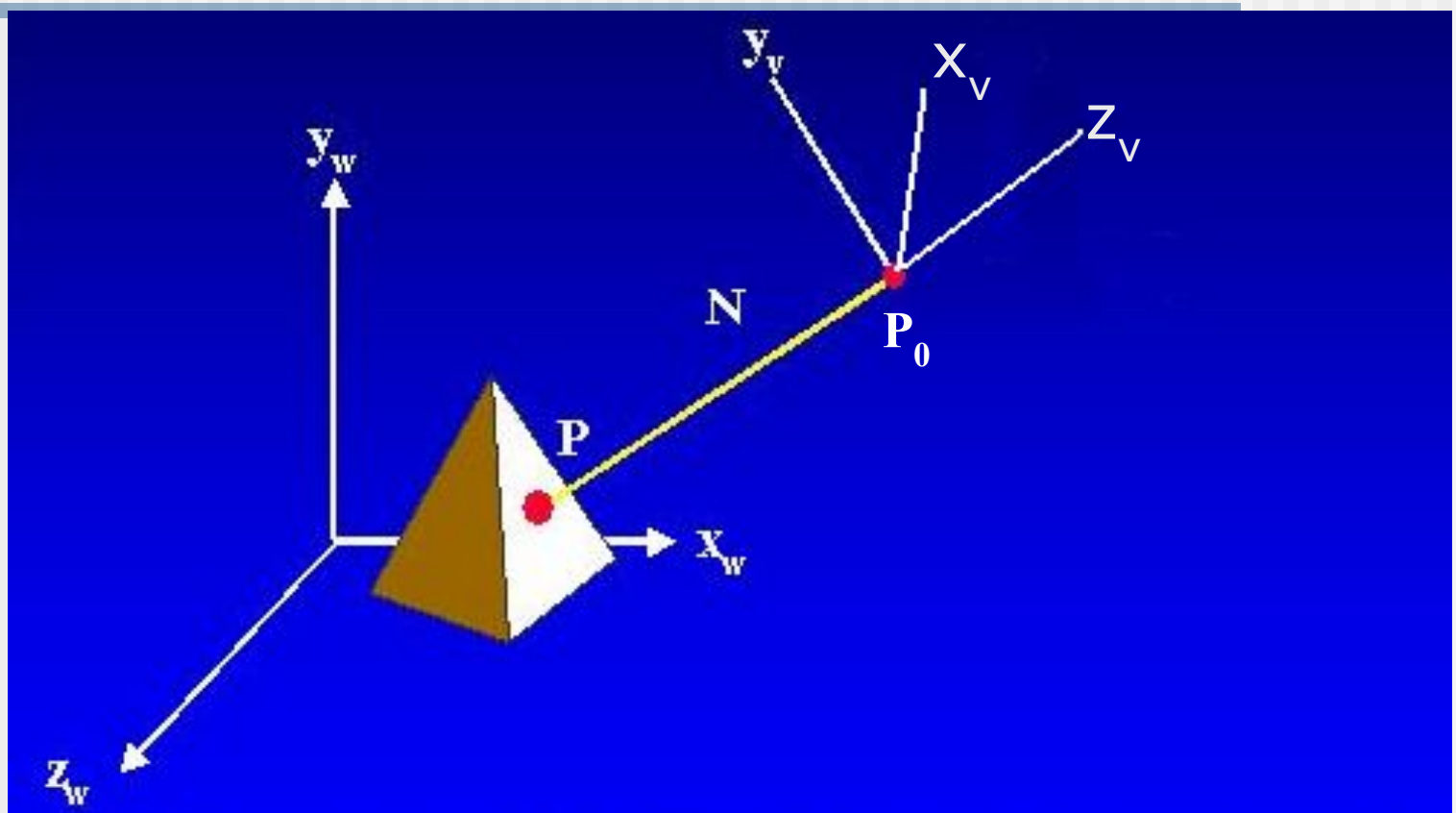
Viewing-Coordinate System

- View plane (or projection plane)
  - Perpendicular to the viewing z_v axis
- View-plane normal vector N
  - Choose a world coordinate position to determine N
  - Determined by a look-at point relative to the view reference point.

Viewing-Coordinate System

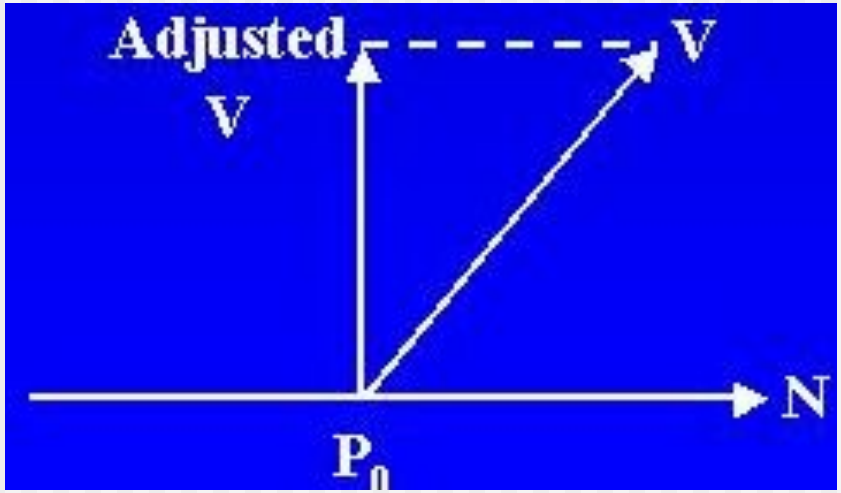


Viewing-Coordinate System



Orientation of the view plane for a specified look-at point P, relative to the viewing-coordinate origin P_0.

Viewing-Coordinate System

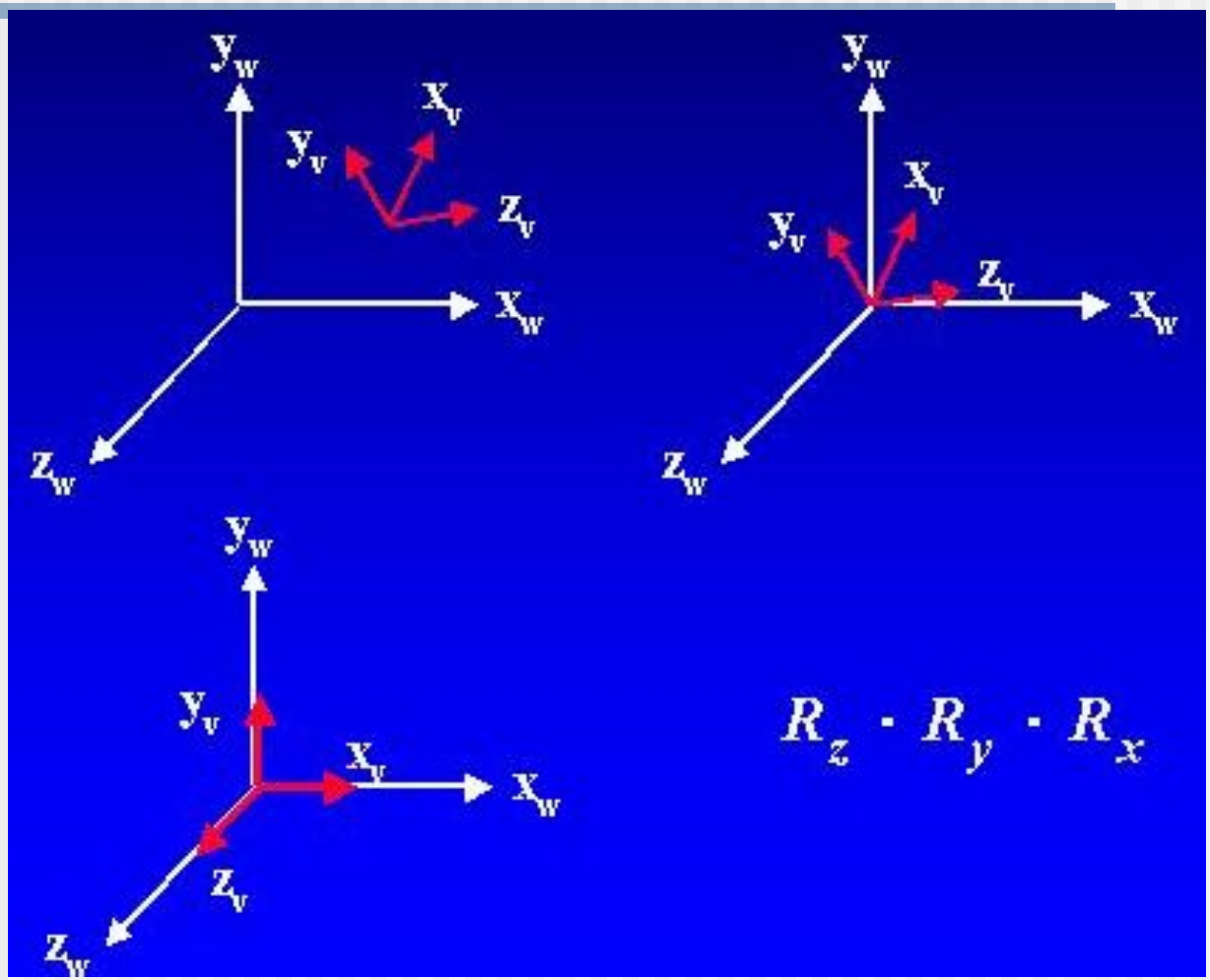


- **View-up vector V**
  - This vector is used to establish the positive direction for the y_v axis.
  - It is difficult to determine the direction for V that is precisely perpendicular to N.
    - V is adjusted so that it is projected into a plane that is perpendicular to the normal vector.

Viewing-Coordinate System

- View-plane distance
  - Choose the position of the view plane along the z_v axis.
  - The view plane is always parallel to the $x_v y_v$ plane.
- Right-handed viewing system
  - The convention of PHIGS and OpenGL
- To obtain a series of views of a scene
  - Fix the view reference point and change the direction of N.
  - The normal vector N is the most often changed viewing parameter

Transformation from World to Viewing Coordinates



Transformation from World to Viewing Coordinates

$$n = \frac{N}{|N|} = (n_1, n_2, n_3)$$

$$u = \frac{V \times N}{|V \times N|} = (u_1, u_2, u_3)$$

$$v = n \times u = (v_1, v_2, v_3)$$

$$R = \begin{bmatrix} u_1 & u_2 & u_3 & 0 \\ v_1 & v_2 & v_3 & 0 \\ n_1 & n_2 & n_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

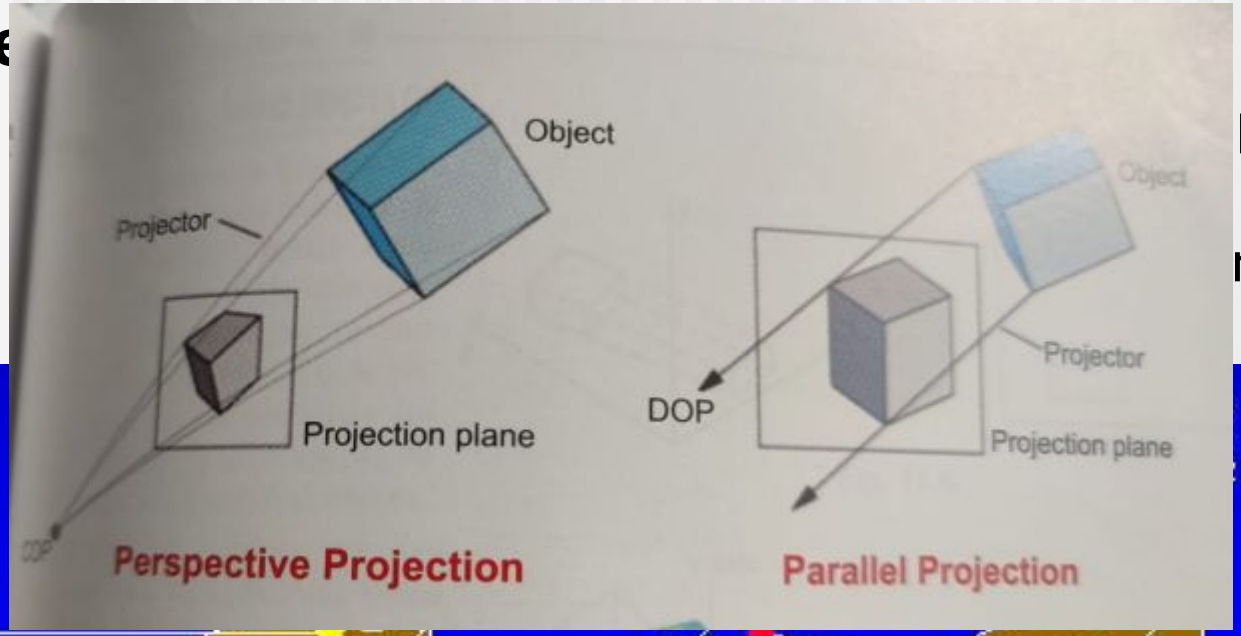
$$M_{WC,\,VC} = R.\ T$$

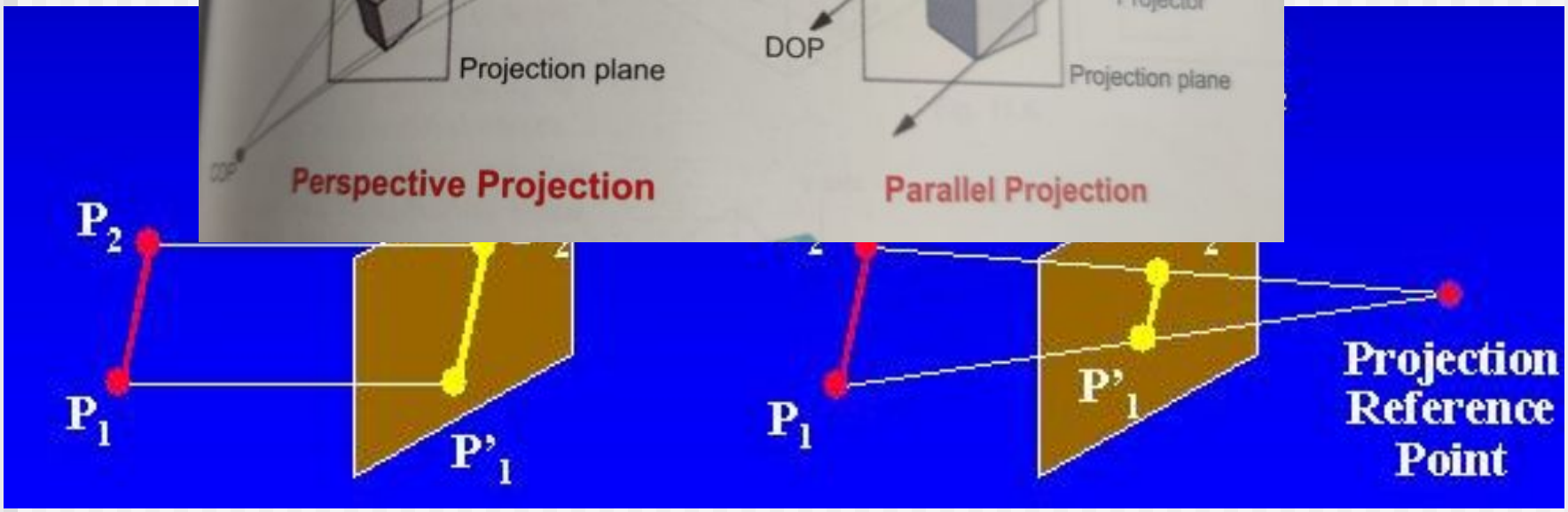
Projection

- **Projection plane (or view plane)**
- **Center of projection (or projection reference point)**
  - An arbitrary point in the three-dimensional space.
  - Usually it is the view point.
- **Projectors**
  - Lines from the center of projection through each point in an object.
- **Parallel projection**
  - The center of projection is located at infinity.
  - All the projectors are parallel

Projection

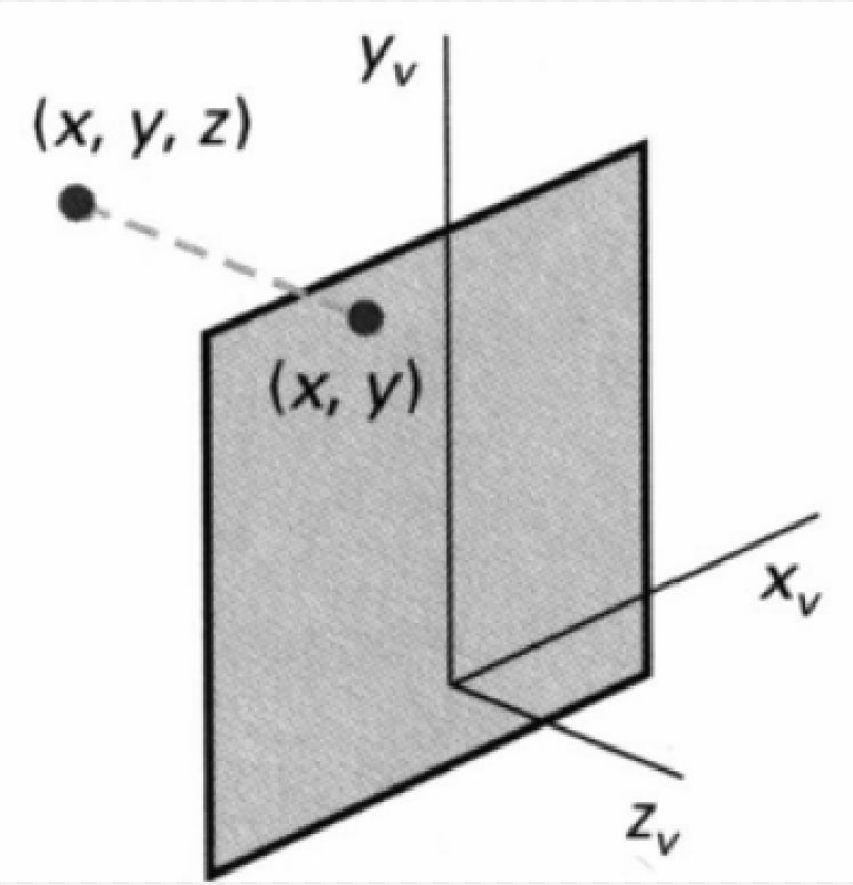
- **Pe**
  - point in
  - rer line of





Orthographic Parallel Projection

Projection is perpendicular to the view plane

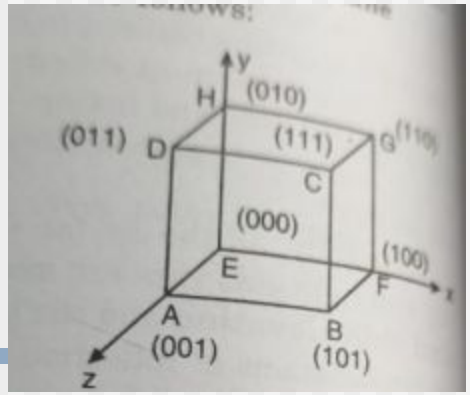


$$x_p = x$$
$$y_p = y$$
$$z_p = 0$$

$$\mathbf{M}_{\text{parallel}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Front, side, rear orthographic projections of an object are called elevations
Top orthographic projection is called a plan view

Finding orthographic projection of a unit cube onto x=0, y=0, z=0 plane.



$$\begin{array}{c} A \\ B \\ C \\ D \\ E \\ F \\ G \\ H \end{array} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Transformation matrix for $x = 0$ plane is $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

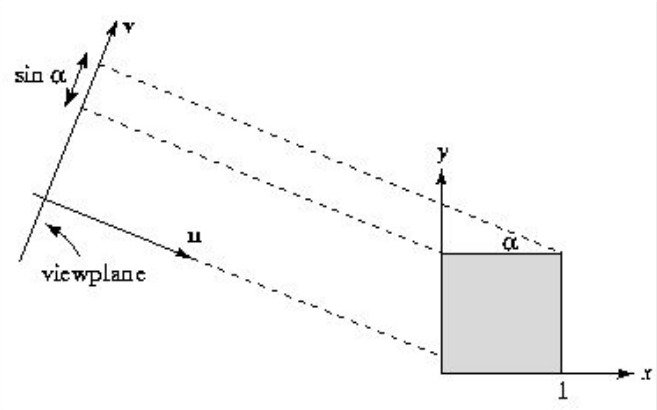
Therefore, the orthographic projection is

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Similarly the orthographic projection onto $y = 0$ plane is

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

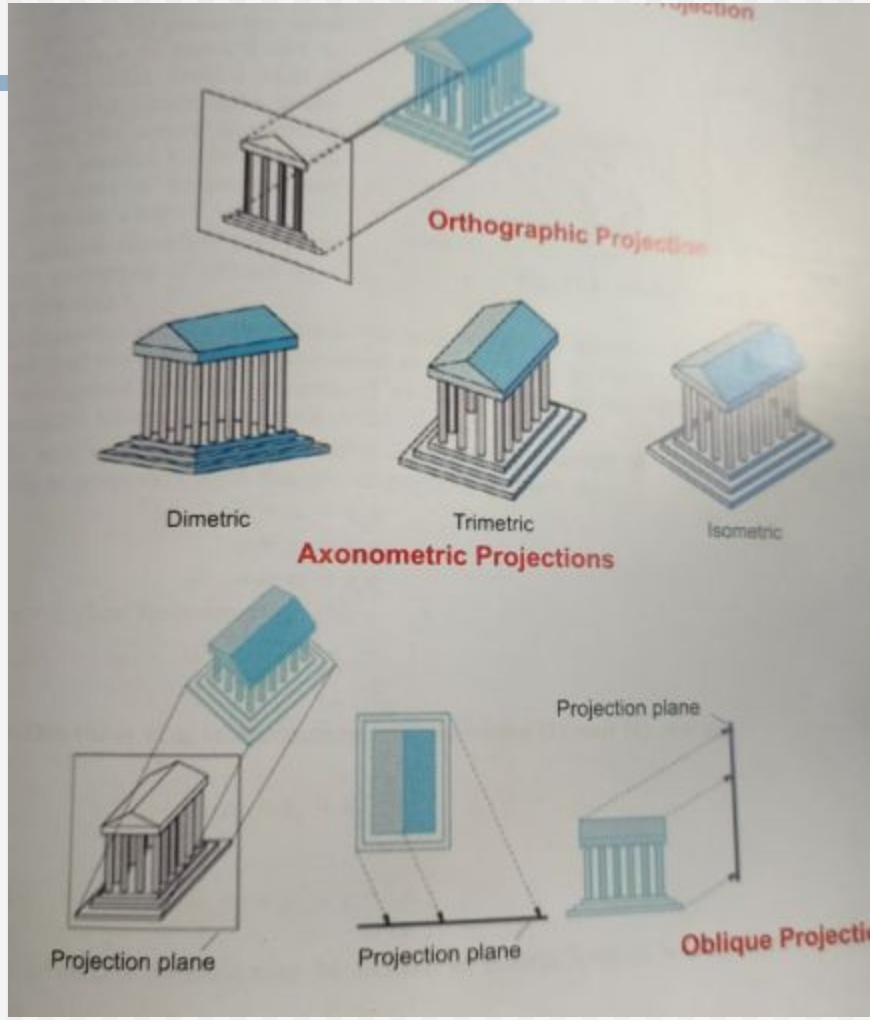
Orthographic Parallel Projection



1. Multiview: Combines multiple view like Top& bottom View, front & rear view etc.
2. Axonometric: Direction of projection is not parallel to any of the 3 principal axis. Constructed using rotation and translation to manipulate the object so that at least 3 adjacent faces are shown. The result id then projected at infinity onto one of the coordinate planes (say z=0) from the center of projection.
   - orthographic projections that display more than one face of an object
   - **n** not parallel to any principal axis
   - Foreshortening factor sin(α) is the ratio of the projected length of a line to its true length.
   - Shows its true shape only when a face is parallel to the plane of projection.
   - Parallel lines are equally foreshortened.

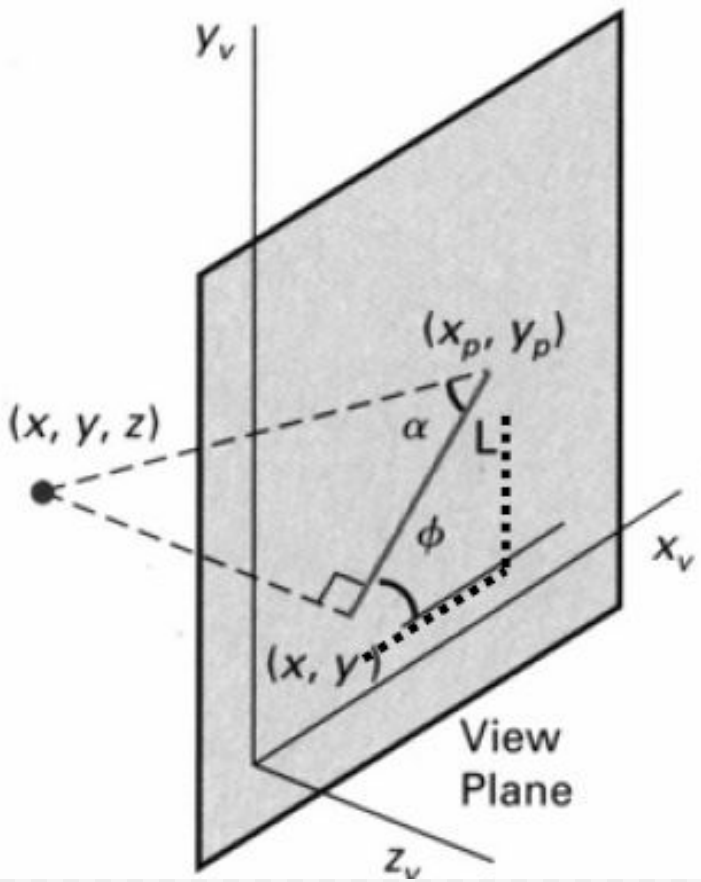
Axonometric **Parallel Projection**

- Isometric: all 3 principal axes foreshortened equally. Direction of projection makes equal angles with all 3 principal axis.
- Dimetric: 2 foreshortened equally. Direction of projection makes equal angles with exactly 2 of the principal axis.
- Trimetric: all 3 foreshortened unequally. Direction of projection makes unequal angles with all 3 principal axis.



Oblique Parallel Projection

Projection is not perpendicular to the view plane



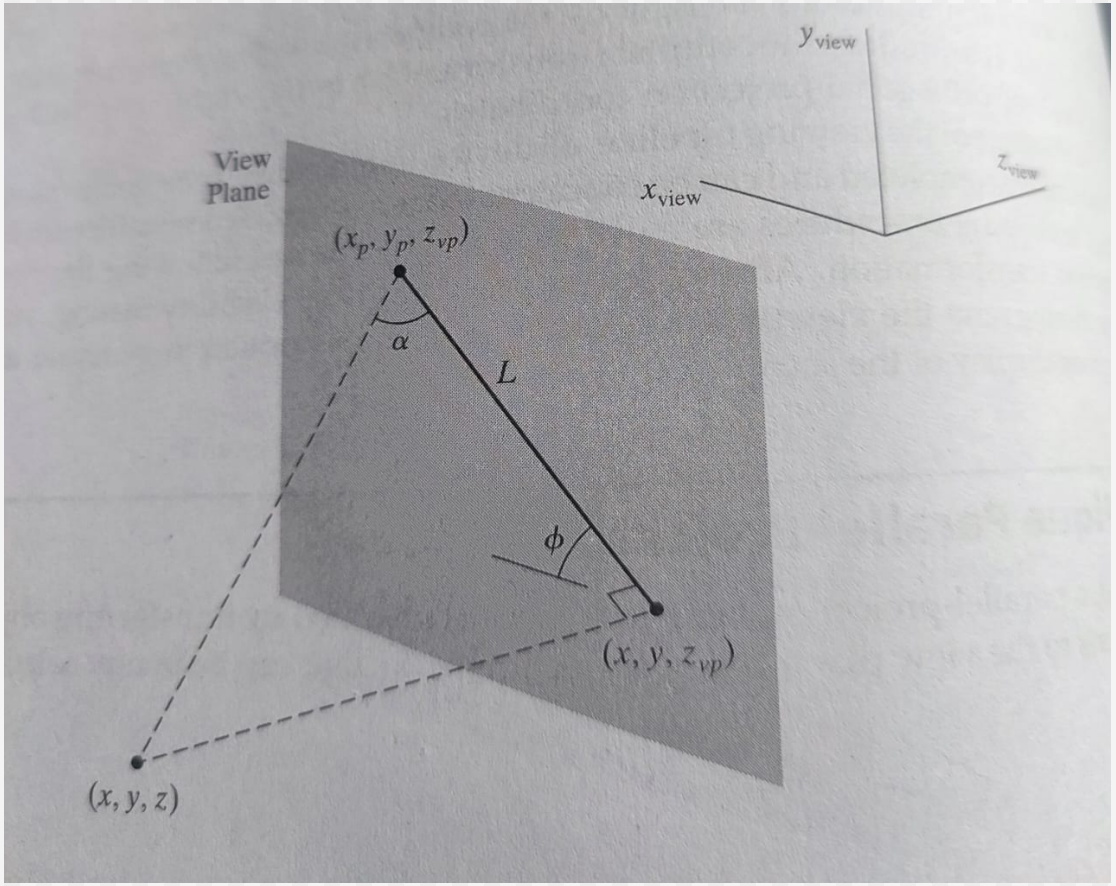
$$\tan\alpha = z/L$$
$$L = z/\tan\alpha$$

$$x_p = x + L\cos\phi$$
$$y_p = y + L\sin\phi$$

$$M_{parallel} = \begin{bmatrix} 1 & 0 & \frac{\cos\phi}{\tan\alpha} & 0 \\ 0 & 1 & \frac{\sin\phi}{\tan\alpha} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

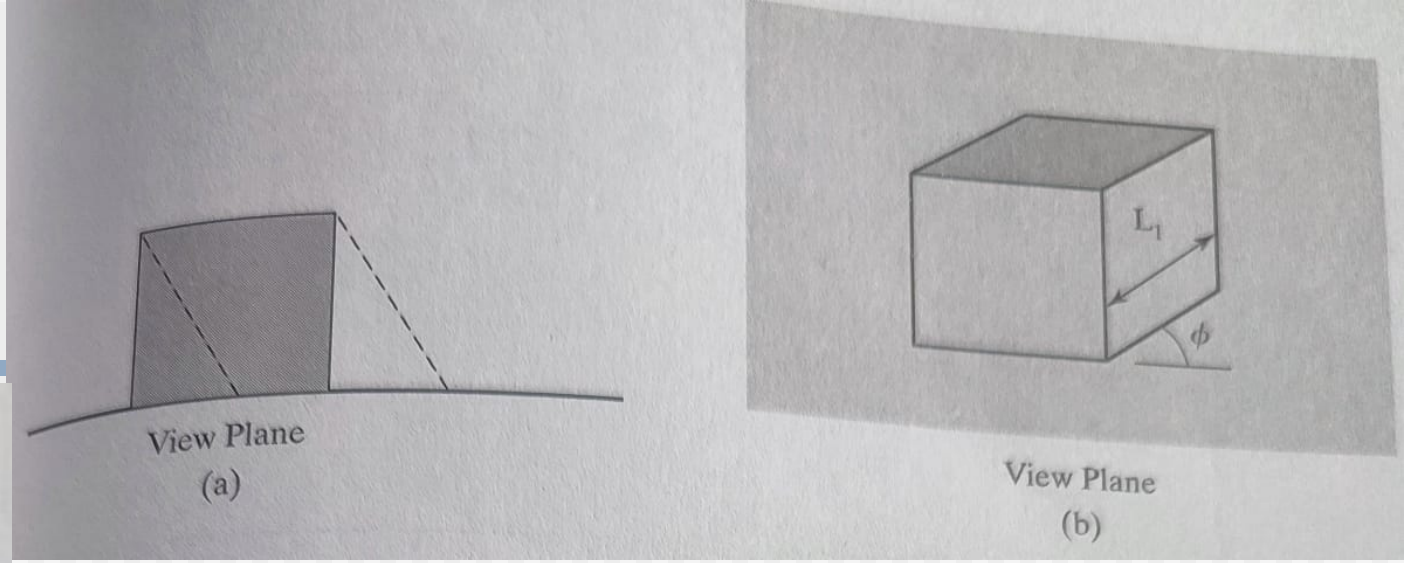
$$x_p = x + L\cos\phi$$
$$y_p = y + L\sin\phi$$

0<=a <=90
0<= φ<=360



$$\tan\alpha = \frac{z_{vp} - z}{L}$$

$$L = \frac{z_{vp} - z}{\tan\alpha} = L_1(z_{vp} - z)$$



$L_1 = \cot\alpha$, which is also the value of L when $z_{vp} - z = 1$

Oblique Parallel projection can be given as

$$x_p = x + L_1(z_{vp} - z)\cos\phi$$

$$y_p = y + L_1(z_{vp} - z)\sin\phi$$

When $L_1 = 0$ ($\alpha = 90^0$ we get orthogonal projection.

An oblique projection is obtained by projecting points along parallel lines that are not perpendicular to the projection plane. Figure shows an oblique projection of a point $P(x, y, z)$ on the $z = 0$ plane.

P_{ob} (x_p, y_p) is oblique projection of the point P and P_{or} (x, y) is the orthographic projection of P on the $z = 0$ plane.

Let L be the length of the line joining points P_{ob} (x_p, y_p) and P_{or} (x, y), θ be the angle made by the line L with x axis.

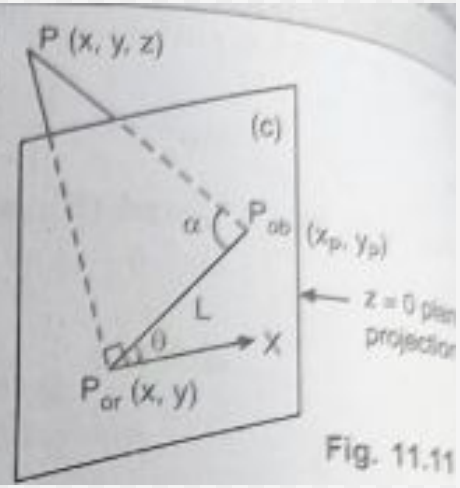


Fig. 11.11

We can express the projection coordinates in terms of x, y, L and θ as

$$x_p = x + L \cos \theta$$
$$y_p = y + L \sin \theta$$

Length L depends on the angle α and the Z coordinate of the point to be projected

$$\tan \alpha = \frac{Z}{L} \Rightarrow L = \frac{Z}{\tan \alpha} = ZL_1 \qquad \left(\text{where } L_1 = \frac{1}{\tan \alpha}\right.$$

when $\quad Z = 1, \; L_1 = L$

Now, we can write the oblique projection as

$$x_p = x + Z(L_1 \cos \theta)$$
$$y_p = x + Z(L_1 \sin \theta)$$

Considering $\quad Z_p = 0$, we get

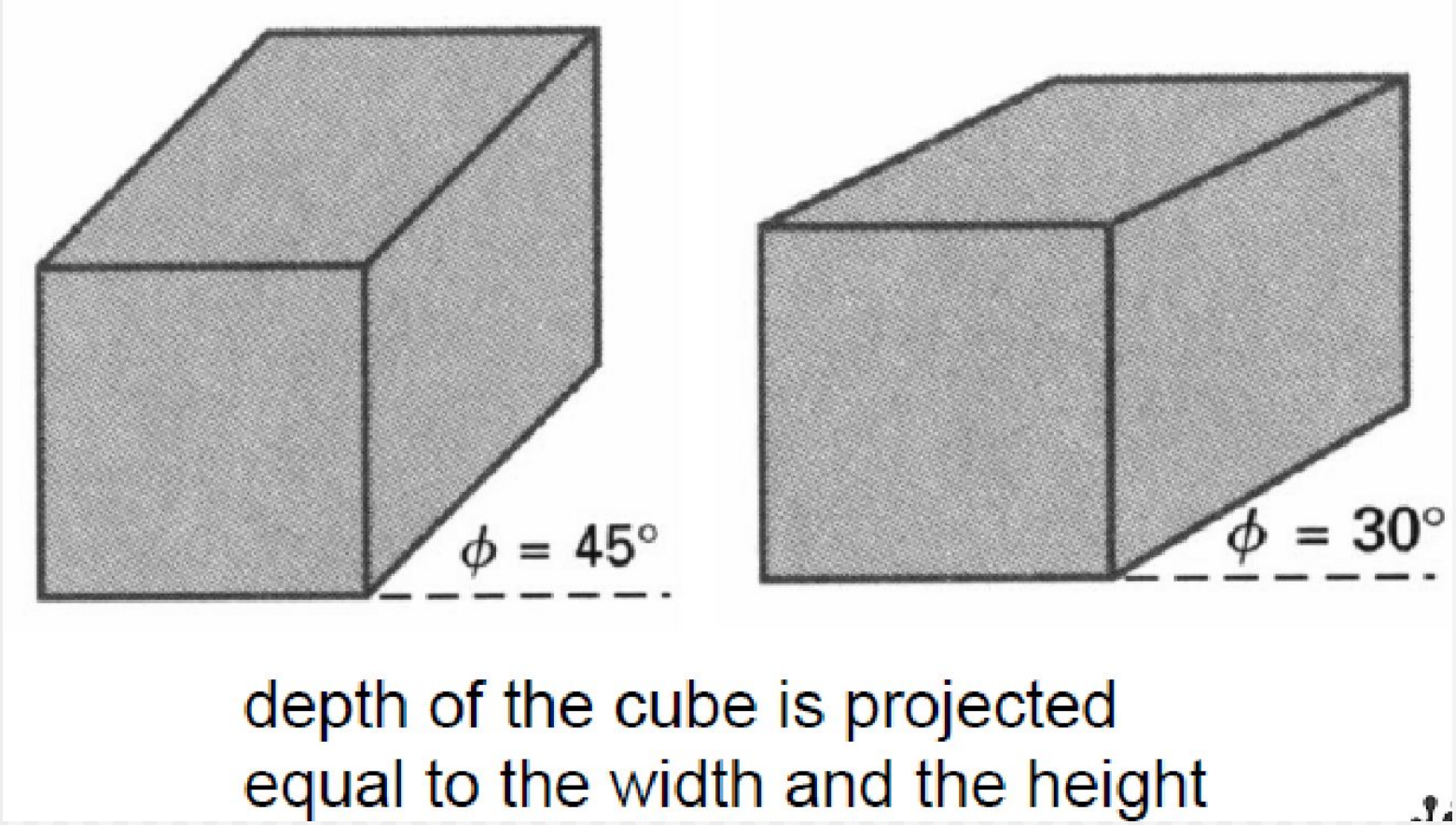
$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & L_1 \cos\theta & 0 \\ 0 & 1 & L_1 \sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

This is the standard matrix for oblique projection onto $z = 0$ plane.

Note: When

Parallel Projection: Cavalier Projections

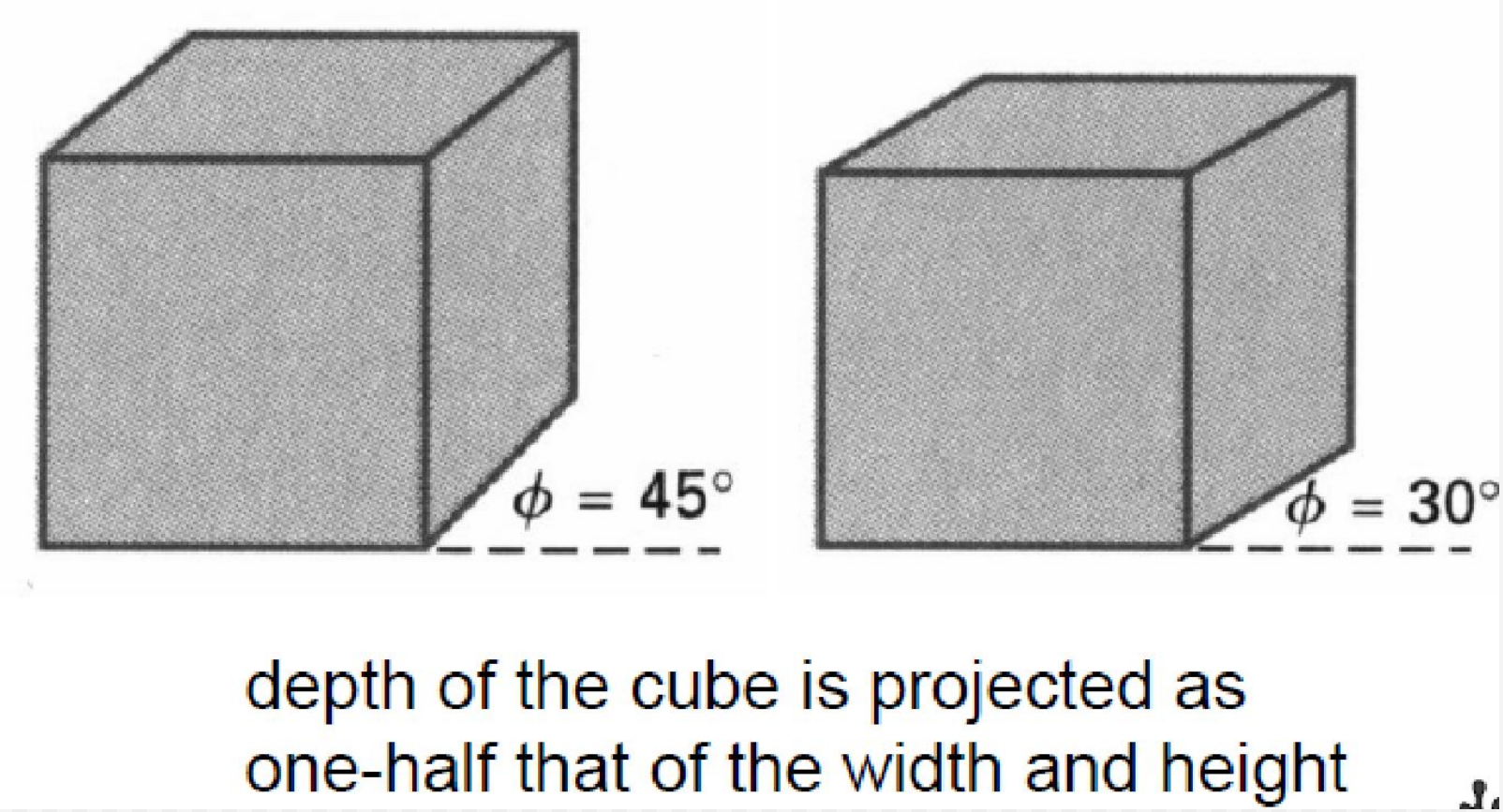
When the angle the oblique projectors and the plane of projection is 45° and the foreshortening factor for all 3 principal axis are equal. Resulting figure is thicker. No foreshortening of lines perpendicular to xy-plane.



Lines perpendicular to the projection plane are projected with no change in length

Parallel Projection: Cabinet Projections

When the angle the oblique projectors and the plane of projection is $\cot^{-1}(1/2)=63.43^{o}$ and the foreshortening factor for edge perpendicular to the plane of projection is 1/2. Corrects distortion produced by cavalier projection.



Lines perpendicular to the projection plane are projected at ½ of their length. They appear more realistic

Find the transformation for a) Cavalier projection with $\theta=45^0$ (b) cabinet projection with $\theta=30^0$ (c) Show the projections for Unit Cube.

a) Cavalier projection is an oblique projection where there is no foreshortening of lines perpendicular to xy plane, hence, L1=f=1, $\theta=45^0$

$$T=\begin{bmatrix} 1 & 0 & f\cos\theta & 0 \\ 0 & 1 & f\sin\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad T=\begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b) Cabinet projection is an oblique projection with lines perpendicular to the projection plane being projected at ½ of their length. Hence, L1=f=1/2, $\theta=30^0$

$$T=\begin{bmatrix} 1 & 0 & \frac{\sqrt{3}}{4} & 0 \\ 0 & 1 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

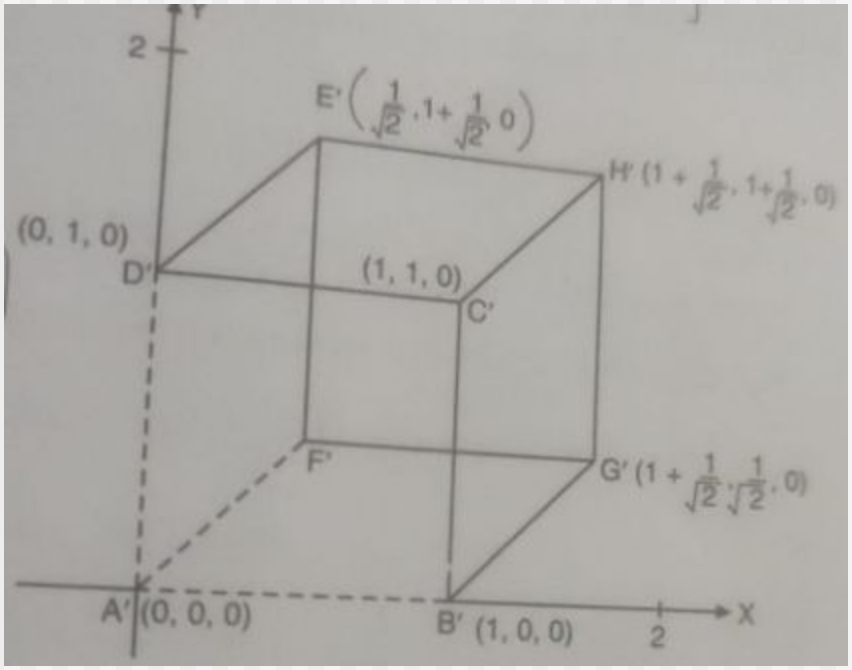
c) Vertices of unit cube for Cavalier projection are

$$U = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$U' = T.U$$

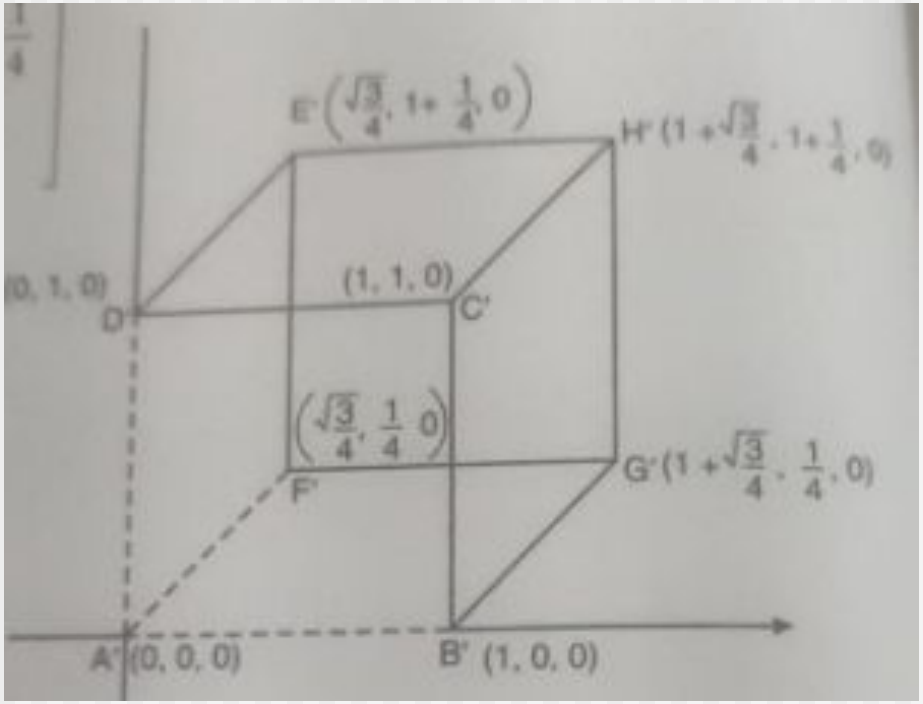
$$U' = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$U' = \begin{bmatrix} 0 & 1 & 1 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1+\frac{1}{\sqrt{2}} & 1+\frac{1}{\sqrt{2}} \\ 0 & 0 & 1 & 1 & 1+\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1+\frac{1}{\sqrt{2}} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

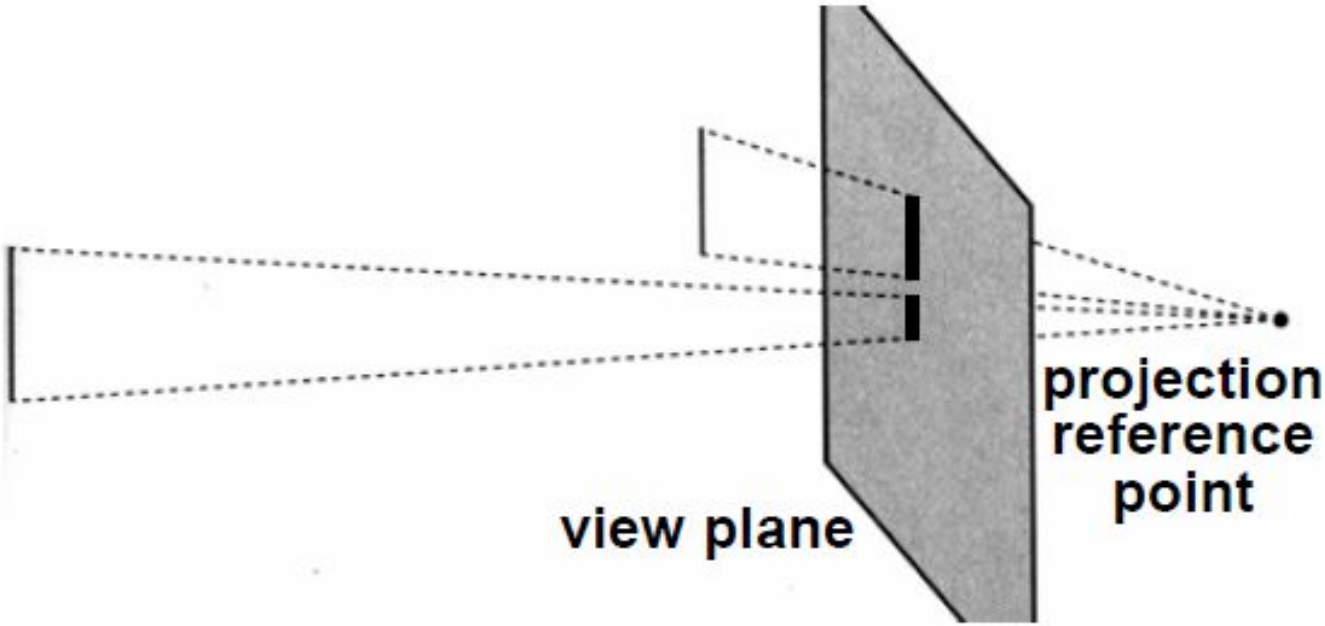


c) Vertices of unit cube for Cabinet projection are

$$U' = \begin{bmatrix} 0 & 1 & 1 & 0 & \frac{\sqrt{3}}{4} & \frac{\sqrt{3}}{4} & 1+\frac{\sqrt{3}}{4} & 1+\frac{\sqrt{3}}{4} \\ 0 & 0 & 1 & 1 & 1+\frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 1+\frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$



Perspective Projection



perspective projection of equal-sized objects at different distances from the view plane

- Vanishing points: Illusion that after projection certain set of parallel lines appear to meet at some point on the projection plane (PP).
- Vanishing points for any set of lines that are parallel to one of the principal axis of an object is referred to as a principal or axis vanishing point.
- Number of principal vanishing points vary from 1 to 3 depending on the number of principal axis intersecting the PP.
- Single point perspective projection: Only one principal axis intersects the PP. Three types of 1-pont perspective projection:
  - Projectors are located at x-axis
  - Projectors located at y-axis
  - Projectors located at z-axis

One-point Perspective projection

- Projectors are located at z-axis

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

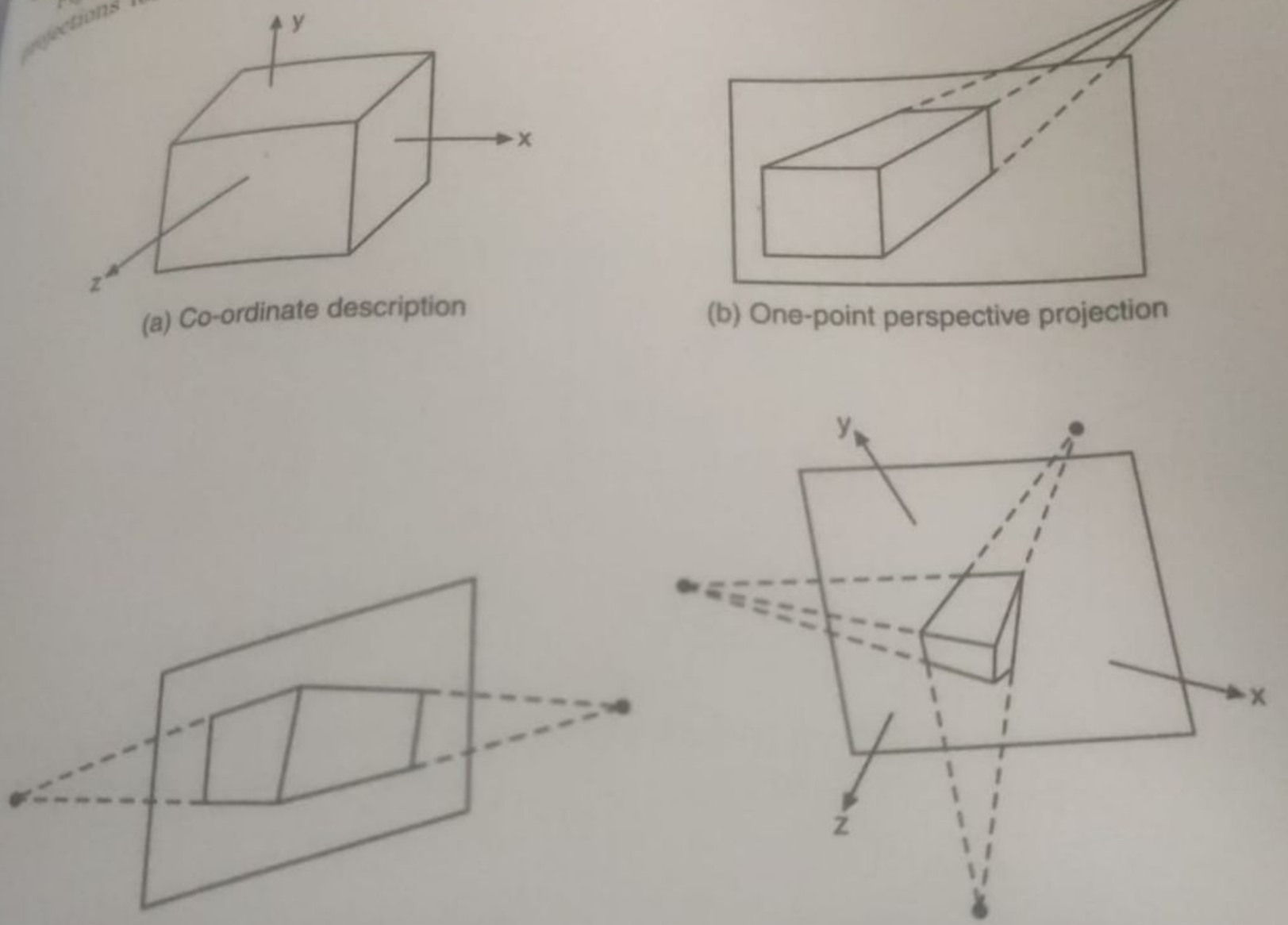
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ rz+1 \end{bmatrix}$$

Where, rz+1 != 1

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{x}{rz+1} \\ \frac{y}{rz+1} \\ \frac{z}{rz+1} \\ 1 \end{bmatrix}$$

It has center of projection at $\begin{bmatrix} 0 \\ 0 \\ -\frac{1}{r} \\ 1 \end{bmatrix}$ and vanishing point on z-axis at $\begin{bmatrix} 0 \\ 0 \\ \frac{1}{r} \\ 1 \end{bmatrix}$

Figure sh
projections for a cube.



(a) Co-ordinate description

(b) One-point perspective projection

(c) Two-point perspective projection

(d) Three-point perspective projection

One-point Perspective projection

- Projectors are located at x-axis

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ px+1 \end{bmatrix}$$

Where, px+1 != 1

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{x}{px+1} \\ \frac{y}{px+1} \\ \frac{z}{px+1} \\ 1 \end{bmatrix}$$

It has center of projection at $\begin{bmatrix} -\frac{1}{p} \\ 0 \\ 0 \\ 1 \end{bmatrix}$ and vanishing point on x-axis at $\begin{bmatrix} \frac{1}{p} \\ 0 \\ 0 \\ 1 \end{bmatrix}$

One-point Perspective projection

- Projectors are located at y-axis

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & q & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

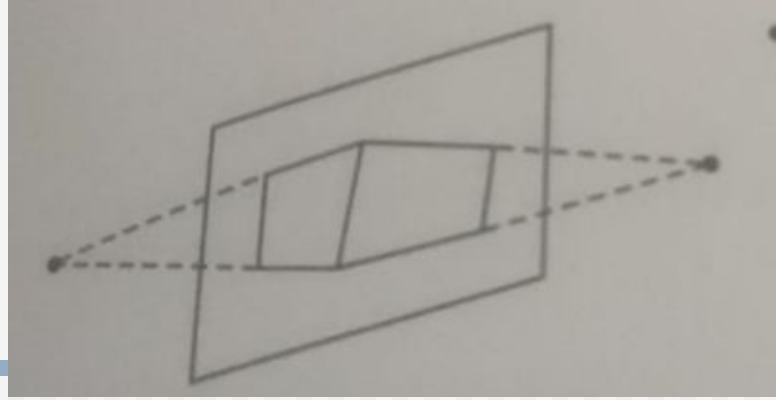
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ qy+1 \end{bmatrix}$$

Where, px+1 != 1

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{x}{qy+1} \\ \frac{y}{qy+1} \\ \frac{z}{qy+1} \\ 1 \end{bmatrix}$$

It has center of projection at $\begin{bmatrix} 0 \\ -\frac{1}{q} \\ 0 \\ 1 \end{bmatrix}$ and vanishing point on x-axis at $\begin{bmatrix} 0 \\ \frac{1}{q} \\ 0 \\ 1 \end{bmatrix}$

Two point perspective projection



- When PP intersects exactly two principal axis.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & q & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ px+qy+1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \dfrac{x}{px+qy+1} \\ \dfrac{y}{px+qy+1} \\ \dfrac{z}{px+qy+1} \\ 1 \end{bmatrix}$$

It has 2 COP, one on x-axis at [-1/p 0 0 1] and the other at [0 -1/q 0 1]

It has 2 vanishing points one at x axis at [1/p 0 0 1] and the other on the y axis at [0 1/q 0 1]

- The 2-point perspective transformation can also be obtained by concatenation of two single point perspective transformations.

Three point perspective projection

When PP intersects all three principal axis i.e., none of the principal axis is parallel to the PP. The need of this transformation is to reconstruct the shape of a 3D object, because to draw a cube it is necessary to know the coordinate of the three faces.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & q & r & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \dfrac{x}{px+qy+rz+1} \\ \dfrac{y}{px+qy+rz+1} \\ \dfrac{z}{px+qy+rz+1} \\ 1 \end{bmatrix}$$

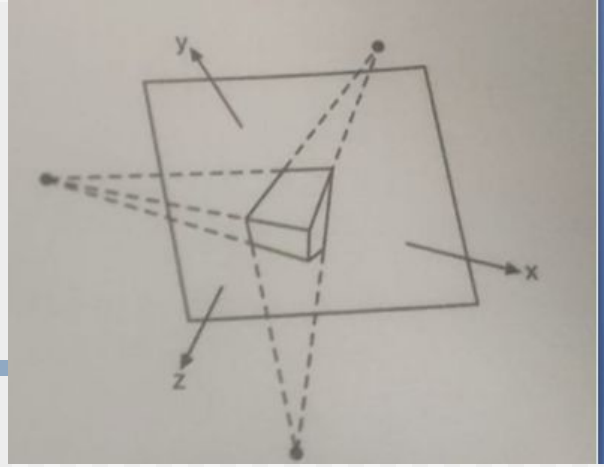
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ px+qy+rz+1 \end{bmatrix}$$

It has 2 COP, one on x-axis at [-1/p 0 0 1] and the other at [0 -1/q 0 1]

It has 2 vanishing points one at x axis at [1/p 0 0 1] and the other on the y axis at [0 1/q 0 1]

- The 2-point perspective transformation can also be obtained by concatenation of

Three point perspective projection



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & q & r & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ px+qy+rz+1 \end{bmatrix}$$

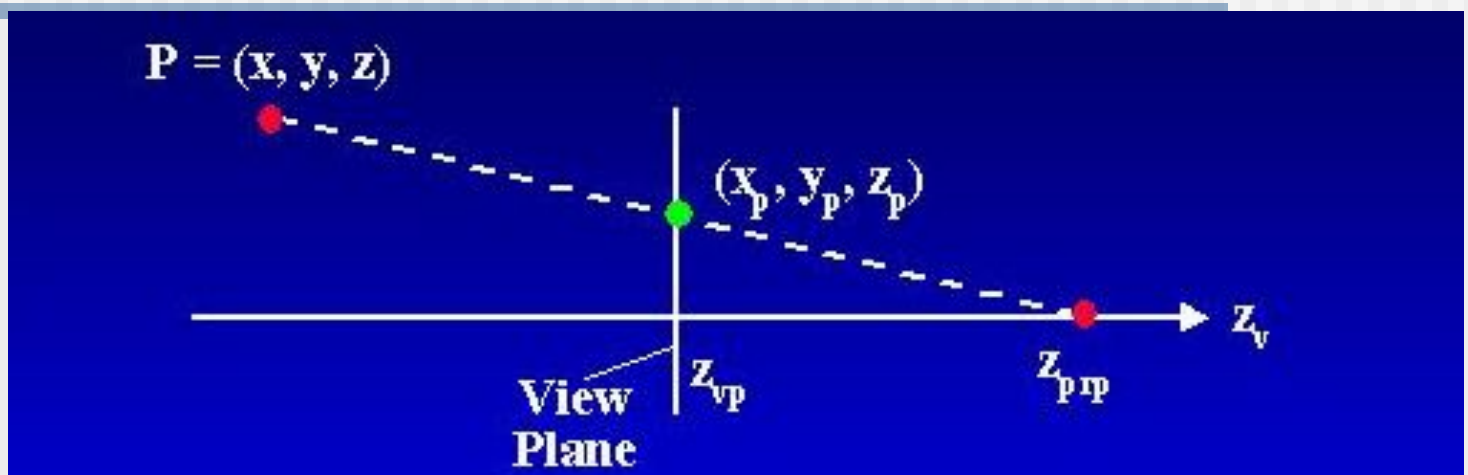
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \dfrac{x}{px+qy+rz+1} \\ \dfrac{y}{px+qy+rz+1} \\ \dfrac{z}{px+qy+rz+1} \\ 1 \end{bmatrix}$$

It has 3 COPs, one on x-axis at [-1/p 0 0 1], second on y-axis at [0 -1/q 0 1] and third on z-axis at [0 0 -1/r 1].

It has 3 vanishing points one on x-axis at [1/p 0 0 1], one on the y axis at [0 1/q 0 1] and the third on z-axis [0 0 0 1/r 1].

- The 3-point perspective transformation can also be obtained by concatenation of three single point perspective transformations.

Perspective Projection



$$x_p = x\left(\frac{z_{prp} - z_{vp}}{z - z_{prp}}\right) = x\left(\frac{d_p}{z - z_{prp}}\right)$$

$$d_p = z_{prp} - z_{vp}$$

$$y_p = y\left(\frac{z_{prp} - z_{vp}}{z - z_{prp}}\right) = y\left(\frac{d_p}{z - z_{prp}}\right)$$

$$z_p = 0$$

Perspective Projection

- **In three-dimensional homogeneous-coordinate representation**

$$\begin{bmatrix} x_h \\ y_h \\ z_h \\ h \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & z_{vp}/d_p & -z_{vp}(z_{prp}/d_p) \\ 0 & 0 & 1/d_p & -z_{prp}/d_p \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$h = (z - z_{prp})/d_p$ is the homogeneous factor

$x_p = x_h / h$ & $y_p = y_h / h$